The Mechanism of Fields from the Bost–Connes System

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Abstract

The structure of the forces and the response of matter to a non-uniform background have been established on the state space of the Bost–Connes system; the field has so far had no definition. Starting from the definition of motion, we define a field as the region-by-region values of a background quantity in the state space; it appears in observational spacetime as the statistics of events. The standing-wave cycles of matter occupy the spectral weight of the region they inhabit and accumulate channel phase along dwell segments, and the corrections are redistributed outward conservatively; the deduction of occupation gives the gravitational potential, the metric, and the source equation, and the accumulation of phase gives the field equations of the electromagnetic, weak, nuclear, and color interactions. Following the duality of discrete and continuous, the equations are written on two levels, discrete structural equations in the state space and differential equations for statistical field quantities in observational spacetime, and the classical Maxwell equations hold closed under the statistical readout. The nonlinear equation of gravity agrees with general relativity at all orders of the existing tests, and the next-order coefficient of the spatial metric component gives a discriminable deviation. The representative puzzles of fields, action at a distance, the photoelectric effect, the localization of field energy, the graviton, and the classical limit, are resolved one by one from the same structure.

Contents

1 The Definition of Fields

1.1 Introduction

This paper is based on the state space established from the Bost–Connes system [15] in reference [1], which supplies the structure of spacetime, matter, and the forces together with all numerical values; the mechanism of thermalization and the carrier of records are established in reference [2]. Within this framework, reference [3] defines the motion of matter. The subject of motion is a stable standing wave maintained by cycles in the thermal bath, and cyclic propagation drives the displacement of the reconstruction center; the record of motion is phase carried away irrevocably by a really emitted photon; the whole process takes place in the state space and is read out as events in observational spacetime; motion thus appears as a sequence of events. Observational spacetime is determined by the observable space at the reference state [1]; it carries events but not processes, and no observer appears in the criteria for a record [3].

This definition takes a single standing wave as its object, and successive events join into its trajectory. Observation contains another kind of appearance. The deflection of a test charge, the bending of light, and the shift of interference fringes are likewise composed of events, yet their content separates into two parts, one belonging to the probe itself and one changing only with position. Replacing the probe at the same position, the force on a test charge scales with its charge and the ratio stays fixed, while the bending is the same for all standing waves, independent of the probe's mass. The positional part already has a general form in the transport equations of reference [3]: when the lattice spacing $\delta(x)$ and the cycle rate $v(x)$ vary slowly with position, all standing waves fall alike, and the slowly varying region is called a gravitational field in observation. Once the probe's share is separated, the remaining reading belongs only to the position; such an appearance is what observation calls a field. The events of a field read out not the doings of any particular piece of matter but the quantity carried by position.

Reference [3] left the identity and origin of the background to the source-side structure of the arena. The source side therefore poses three questions.

- First, the quantity is read out by the probe but not carried by it; it belongs to the background that carries motion. The quantities of reference [1] are all defined on the whole space – what quantity is read out at each position?
- Second, the transport equations take $\delta(x)$ and $v(x)$ as a given background – how does the presence of matter change the values of the background where it sits?
- Third, the background and its values are both in the state space, and observational spacetime carries only events – how are the position-by-position values read out through events?

1.2 The Background Quantities of Fields

The quantity read out at each position belongs to the background that carries motion (§1.1) and is given by the structure of the vacuum. The vacuum is the thermal background: all excitations run on one and the same state space, the state is the KMS state [28], and the state space is the $L^2(\mathbb{Z}/7^{41}\mathbb{Z})$ identified in reference [1]; the Hamiltonian $H = \log n$ (with n the mode label) gives at once the energy spectrum, the spatial spacing, and the time period, the three sharing the same spacing structure, and the thermal weights are $w_\beta(n) = n^{-\beta}/Z(\beta)$, with $Z(\beta) = \sum_n n^{-\beta}$ the partition function.

The fluctuations of the background are measured by the spacing structure. Each mode carries one share of fluctuation $|\nabla H(n)|^2$ according to its nearest-neighbor spacing (the sum of squares of the adjacent lattice spacings), and the fluctuations of all modes accumulate to the intrinsic constant $C^* = \sum_n |\nabla H(n)|^2 = 1.9543783$. For a process of spectral weight W, the characteristic energy $E(W)$ is given by the matching condition: the process side and the background side count one and the same share of fluctuation,

$$E(W)^2\, C^* = W \log^2 W, \tag{1}$$

that is, $E(W) = \log W\, \sqrt{W/C^*}$; units follow the convention of reference [1], with energy in GeV. The vacuum thereby carries two uniform quantities.

One is the effective spectral weight of the condensation. The vacuum runs the electroweak condensation, whose participating set is all $\varphi(7^4) = 2058$ modes of the unit group (φ the Euler function); weak-force virtual processes occupy $N_W^{\rm eff} = 18.25$ shares of it (the effective channel number [1]), and the remaining spectral weight determines the energy scale of the condensation:

$$N_{\rm vac} = \varphi(7^4) - N_W^{\rm eff} = 2039.75, \qquad v_0 = E(N_{\rm vac}) = 246.19\,\text{GeV}, \tag{2}$$

v_0 is the cycle rate, shared by all excitations [3].

The other is the phase of each character channel. The insertion flow of virtual processes accumulates phase along dwell segments, one share of Born-level coupling per insertion [1]; the insertion flow of the vacuum is the same everywhere, so the channel phases are uniform, and their value is set to zero below.

The two quantities are the two parts of the polar decomposition of one and the same correction. The insertion flow only accumulates phase without changing the modulus, and the correction of the modulus is already borne by the effective channel number N^{eff} [1]. The polar decomposition $S = J\Delta^{1/2}$ of the Tomita operator [29] is unique: $\Delta^{1/2}$ is the positive-semidefinite modular part, and J is the antilinear involution [1]. The correction of the modulus falls on $\Delta^{1/2}$, and the correction of the phase falls on J; the assignment of the two halves is fixed by the uniqueness of the decomposition. Both quantities take uniform values on the whole space. The non-uniform part is brought by matter.

Matter is a stable standing wave maintained by cycles in the thermal bath [1]; the standing wave lets amplitude flow out into the vacuum to maintain its formation and retrieves it in detailed balance [3]. The exchange of the cycle requires carrying from the background, one share of fluctuation carrying one exchange; the exchange proceeds station by station along character channels [1]; these channels are exactly the participating set of the condensation, and a share in use by a cycle is not simultaneously available to the condensation, so the occupation is deducted from the spectral weight of the condensation.

Non-uniformity arises from the position of matter. The position of a standing wave is given by motion, the cycle-by-cycle displacement of the reconstruction center [3]; the cycles take place only in the region where the standing wave sits, so the consumption of carrying and the accumulation of phase fall only in that region. The occupation of channels by weak-force virtual processes is the same deduction operation: virtual processes are distributed uniformly over the whole space and the background stays uniform; the cycles of a standing wave concentrate in its region, the background where matter sits acquires a deviation, and the two quantities are no longer uniform. The quantitative form of the deduction is not taken from this correspondence; it is locked independently by additivity in §2.1.

The correction of matter to the background splits into two halves, matching the polar decomposition of the vacuum layer half by half.

The cycles of the standing wave occupy the spectral weight of the region they sit in. The carrying consumed by the cycles is drawn from that region, and the occupied shares are deducted from the spectral weight available to the region: writing $n_{\text{occ}}(U)$ for the occupied shares in region U, the effective spectral weight of the region is

$$N_{\text{vac}}(U) \;=\; N_{\text{vac}} - n_{\text{occ}}(U), \tag{3}$$

the relation between n_{occ} and the energy of the standing wave is fixed in §2.1.

A standing wave carrying character charge accumulates the channel phase of its region. The accumulation proceeds along dwell segments, and the phase of channel k in region U deviates from the vacuum value by one local quantity:

$$a_k(U) \;=\; \big(\text{phase value of channel } k \text{ in } U\big) - \big(\text{vacuum value}\big), \tag{4}$$

the vacuum value is the uniform phase above, and once it is set to zero, $a_k(U)$ is the local reading of the channel phase.

In the continuous coordinates of the readout the two corrections are written as fields of position, $n_{\text{occ}}(x)$ and $a_k(x)$, and the effective spectral weight of a region becomes $W(x) = N_{\text{vac}} - n_{\text{occ}}(x)$; the construction of regions and position readings is given in §1.4. The uniform part is already borne by the vacuum layer (N_W^{eff} and the vacuum phase); $n_{\text{occ}}(x)$ and $a_k(x)$ count only the non-uniform part brought by matter; far from matter both vanish, and the results of the uniform vacuum are recovered in full.

Localization obeys one rule: process-universal numbers take no part in localization. N_{vac}, C^*, and all coupling constants are fixed by the carrying structure of the processes [1] and do not change with position; every region runs one and the same condensation process, and the participating spectral weight is the same N_{vac}. The measure $\log^2 W$ of each share of fluctuation in the matching condition belongs to this same carrying structure, so localization takes its vacuum value, and the local energy-scale relation is

$$v(x)^2\, C^* \;=\; W(x)\, \log^2 N_{\text{vac}}, \tag{5}$$

$v(x)$ is the cycle rate of the region, whose value is fixed in §2.1. Dividing (5) by the vacuum matching condition $v_0^2\, C^* = N_{\text{vac}} \log^2 N_{\text{vac}}$ gives $(v(x)/v_0)^2 = W(x)/N_{\text{vac}}$, which is the time metric component g_{00} of §2.4. This choice is independently forced by observation: if the measure varied with the local weight, the second-order coefficient of g_{00} would become 2.37, contradicting the perihelion precession (Appendix A). Only the correction quantities take part in localization: occupation, phase, and their redistribution.

1.3 Localization of the Values

The change matter makes to the background where it sits has been written as two local correction quantities (§1.2); the corrections of reference [1] are established on the whole space, and their region-by-region accounting requires an argument, given in three steps.

Step 1: the cyclic exchange of a standing wave proceeds site by site, and pattern reconstruction is completed where the standing wave sits [3]; the propagation of influence is constrained by the finite propagation speed: the path amplitude reaching metric distance D within ℓ cycles satisfies (with β the inverse temperature)

$$|A(D,\ell)| \;\le\; e^{-\beta D}\, N(\ell, D), \qquad N(\ell, D) \le 2^{\ell}, \tag{6}$$

$N(\ell, D)$ being the number of paths reaching distance D within ℓ cycles; for $D > \ell \ln 2/\beta$ the decay of the weight outruns the growth of the count, the amplitude is exponentially suppressed, and outside the light cone there is only an exponential tail [3]. The bound is a statement within a finite number of cycles, and it excludes direct corrections across a distance: the direct correction of matter to the background falls only in the region where it sits, and its influence on distant places must arrive by propagation, whose form is established in §2.2.

Step 2: the matching condition (1) is a share-by-share identity: the process side and the background side count one and the same share of fluctuation [1], and the global equality is the result of summing over all shares. Each share of fluctuation has exactly one position, the position of the event of the process that produces it (Step 3); grouping the sum by position, each share is counted on both sides in the region where it sits, and the grouped equalities hold group by group:

$$\sum_{\text{share}\in U} \big(\text{process side}\big) \;=\; \sum_{\text{share}\in U} \big(\text{background side}\big)\,. \tag{7}$$

What closes region by region is the balance itself, that is, the two ways of counting each share of fluctuation agree within the same region; this is a direct consequence of the assignment of shares and does not rest on the propagation bound. The carrying correlations are established across regions, and their steady-state shape is given by the chain of propagation: each step of the chain is a nearest-neighbor transfer, and the steps superpose into a long-range distribution (§2.2); the field is not truncated by the region-by-region nature of the balance.

Step 3: the two sides of the matching condition are two ways of counting one and the same share of fluctuation [1], and one share of fluctuation has exactly one position, the position of the process that produces it, carried by the readings of events [3]; modes and channels are units of accounting and carry no position. The precedent of reference [1] is exactly of this kind: the N_W^{eff} channels occupied by weak-force virtual processes have no position and belong uniformly to the whole space. Position and energy level are one and the same spacing structure (the three metrics are unified [1]), and the carrying operator of the fluctuation weights is diagonal in the site basis:

$$\Pi_{\text{grav}} \;=\; \sum_n |\nabla H(n)|^2 \, |n\rangle\langle n|, \tag{8}$$

the spectral value is the position reading; the distinction of direction does not belong to the carrying layer and is carried by the readout of events. "A share belongs to region U" therefore means: the event of the process producing that share of fluctuation falls in U, and the two ways of counting share one and the same assignment. The partition into regions is constructed in the next subsection.

The global corrections of reference [1] therefore hold region by region in the steady state, and the background structure at region x is obtained from the vacuum structure by replacing the two uniform quantities with local ones,

$$\boxed{N_{\text{vac}} \;\longrightarrow\; N_{\text{vac}} - n_{\text{occ}}(x), \qquad a_k \;\longrightarrow\; a_k(x).} \tag{9}$$

The domain of validity follows: the source equations are steady-state statements; sources varying slowly in time are treated in the adiabatic approximation, with an error given by the ratio of the rate of change of the source to the relaxation rate of the background [2]; dynamical propagation is established in §2.6; the running of couplings with energy scale (the β evolution of reference [1]) takes no part in the spatial distribution, and the field equations work at the fixed physical energy scale.

1.4 The Statistics of Events

The position-by-position values are in the state space, and observational spacetime carries only events (§1.1); the readout is given by the statistics of events, and the unit of the statistics is the region defined by event readings. The readings of one event in observational spacetime are a spectral value and a direction: the position reading takes the spectral value, and the landing point of an event falls on exactly one site each time; the distinction of positions is carried by the direction of the scattered photon, and the statistics of directions recovers the distribution of landing points [3]. Regions are accordingly defined by spectral value and direction: writing r for the spectral value and $\hat{n}$ for the photon direction,

$$U(r,\hat{n}) \;=\; \big\{\,\text{event } e:\; r_e \in [r, r+\delta),\; \hat{n}_e \in \Omega\,\big\}, \tag{10}$$

where δ is the lattice spacing and Ω the solid-angle element at $\hat{n}$. The spectral value is discrete while the direction is continuous; the duality of discrete and continuous reappears in this division of labor between them [3]. The corrections of matter are credited to the region where their events fall.

The legitimacy of the partition is guaranteed by the event density. The positions of shares are carried by event readings (§1.3), and all the partition requires is that events be assignable to regions: each site offers one event slot per cycle, the number of events in a macroscopic region far exceeds what the partition needs to resolve, and subdivision by direction is unrestricted; the modes of the carrying layer need not be apportioned among regions: modes are units of accounting and carry no position (§1.3). The partition is covariant under change of reference frame. It is defined by event readings; a change of reference frame (translation of the origin, rotation, and uniform motion) acts on the event statistics of the readout layer and is a symmetry inherited kinematically [3]. The readings transform accordingly, the counts are unchanged, and the crediting of the corrections is covariant.

The equations are written on two levels, each on one side of the duality of discrete and continuous [3, Appendix A (The Duality of Discrete and Continuous)]. The equations of the state space are discrete structural equations: the fields take values region by region, the conditions of conservation, balance, and propagation have sites and regions as their domain, and the discrete Poisson identity of reference [1] is a precedent of this kind. The equations of observational spacetime are differential equations: the coordinates take values in the completion of the set of readings, single readings fall in a countable dense set, the numerical values of statistics require the real field, and the completion is unique up to isometry [3]; statistical readings become continuous field quantities on the geometry of the completion, and all subsequent differential equations belong to observational spacetime. The conversion between the two levels is given by the truncation of differences against derivatives,

$$f(x+\delta)+f(x-\delta)-2f(x) \;=\; \delta^2 f''(x)+O\big(\delta^4\big), \tag{11}$$

with relative error of order $(\delta/L)^2$ (L the scale of variation of the field); the lattice spacing is one step of the cycle, $1/v$, the minimal step of position readings [3], and at the scale of the solar radius the error is about 10^{-54}, below all experimental precision; the discrete structure leaves testable quantities in the readout: the deviation of the lattice dispersion and the exponential tail outside the light cone [3]. The place where the continuous approximation fails, that is, where the scale of variation of the field shrinks to the order of the lattice spacing, coincides with the boundary of validity of localization (§2.5).

1.5 The Basic Units of Fields

The basic units of fields arise from the basic unit of motion and likewise fall into three processes. The basic unit of motion is the single-cycle propagation of a standing wave, each cycle consisting of free evolution, formation exchange, and pattern reconstruction [3]; the shares of a field are produced in the two stages of exchange and dwell, and become a distribution through a third process,

$$\big(W^{(n)},\, a_k^{(n)}\big) \;\xrightarrow{\text{occupation}}\; \cdot \;\xrightarrow{\text{accumulation}}\; \cdot \;\xrightarrow{\text{propagation}}\; \big(W^{(n+1)},\, a_k^{(n+1)}\big)$$

(superscripts label the cycle number): one cycle of motion takes the standing wave from $(\psi^{(n)}, x_n)$ to $(\psi^{(n+1)}, x_{n+1})$ [3], and the same cycle takes the background quantities of the region from $(W^{(n)}, a_k^{(n)})$ to $(W^{(n+1)}, a_k^{(n+1)})$.

First, occupation is given by formation exchange. Formation exchange requires carrying from the background: the exchange is a single cyclic process of outflow and retrieval [1], proceeding station by station along character channels, one share of fluctuation carrying one exchange; the shares in use are unavailable to the condensation during the cycle, and occupation is the deduction from the spectral weight of the region. The phase per cycle E/v splits into the translation share and the exchange share [3],

$$\frac{E}{v} \;=\; \frac{p^2}{vE} \;+\; \frac{m^2}{vE}\,, \qquad \text{at rest the exchange share is } \frac{m}{v}\,, \tag{12}$$

the number of occupied shares is proportional to the amount of exchange phase, and summing over the region gives $n_{\mathrm{occ}}(U)$ (§1.2; the derivation is in §2.1).

Second, accumulation is given by the insertion flow of virtual processes. The insertion flow proceeds along dwell segments, one share of Born-level coupling phase per insertion [1],

$$\Delta a_k\big|_{\text{per insertion}} \;=\; \alpha_k^{(0)}\,, \tag{13}$$

$\alpha_k^{(0)}$ being the Born-level coupling of the channel [1]; the shares are credited to the channel phase of the region where the charge-carrying standing wave sits, accumulating to $a_k(U)$ (§1.2); the insertion flow only accumulates phase without changing the modulus, falling on the involution part of the polar decomposition.

Third, propagation is given by nearest-neighbor transfer. Matter consumes one share of carrying per cycle, and the consumption is resupplied site by site by nearest-neighbor transfer: each step spans one site, the proper step is $1/v$, and the rate is the cycle rate v,

$$\text{diffusion coefficient} \;\propto\; \left(\frac{1}{v}\right)^2 v \;=\; \frac{1}{v}\,, \tag{14}$$

in the steady state the flux is constant and the distribution stationary, and the distribution of the field is maintained by this steady transport (§2.2; the proper construction is used in §2.5).

The basic units of the three processes are collected as:

Process	Unit	Amount per share	Output
Occupation	one share of fluctuation carries one exchange	exchange share $m^2/(vE)$	$n_{\mathrm{occ}}(U)$, spectral weight field
Accumulation	one share of phase per insertion	$\alpha_k^{(0)}$	$a_k(U)$, channel phase field
Propagation	one site per step	diffusion coefficient $\propto 1/v$	steady distribution (§2.2)

The spectral weight field is the region-by-region count of occupied shares, the channel phase field is the region-by-region accumulation of phase, and the values update cycle by cycle; motion is the cycle-by-cycle displacement of the reconstruction center, and a field is the cycle-by-cycle crediting and propagation of shares.

1.6 The Definition of Fields

The definition of fields extends the definition of motion (§1.1). Fields and motion come from one and the same chain, the same cycles, the same records, the same events; the fork lies only in how the events are read. Events read one by one in order of succession join into the trajectory of matter, which is the appearance of motion; events merged into statistics by position recover the distribution of the background, which is the appearance of a field.

The mechanism of fields consists of four elements.

- **Background** – the carrying of fields is the two uniform quantities of the vacuum: the spectral weight of the condensation and the phase of each character channel, the two parts of the polar decomposition of one and the same correction (§1.2).

- **Source** – the source of fields is the standing-wave cycling of matter, which occupies the spectral weight and accumulates the channel phase of the region it sits in; the corrections are credited only to the region where the cycles take place (§1.2).

- **Propagation** – the local corrections extend by propagation into a position-by-position distribution, and the steady-state shape is the zero-frequency section of propagation (§2.2).

- **Statistics** – the field in observation is the readout of the position-by-position background quantities through the statistics of events.

The complete structure of fields is

$$
\begin{array}{rc}
\text{state space} & \boxed{\text{standing-wave cycles} \xrightarrow{\text{occupation, accumulation}} n_{\text{occ}},\ a_k \xrightarrow{\text{propagation}} n_{\text{occ}}(x),\ a_k(x)} \\
& \wr\ \ \text{event statistics} \\
\text{observational spacetime} & \boxed{\Phi,\ g_{\mu\nu}\ \big|\ \vec{E},\ \vec{B}}
\end{array}
$$

The upper box lies entirely in the state space, and the two arrows are the three processes of the basic units (§1.5): the cycles of the standing wave give the local corrections through occupation and accumulation, and the corrections become the position-by-position background distribution through propagation; the lower box holds the field quantities of observational spacetime, the potential and the metric for gravity and the field strengths for the channels, the two parts separated by the vertical bar, which are the two halves of the polar decomposition.

Matter makes the two background quantities non-uniform, each region having its own values; this family of region-by-region values is the field, and fields live in the state space together with motion. In the continuous coordinates of the readout the fields are written as functions of position,

$$W(x) \;=\; N_{\text{vac}} - n_{\text{occ}}(x), \qquad a_k(x), \quad k \text{ ranging over the resonance channels}\,, \tag{15}$$

W is the spectral weight field and a_k the channel phase fields. The content of the two fields is the state of the background: $W(x)$ is the share of spectral weight available to the condensation in region x, and $a_k(x)$ is the phase value of channel k there. The values sit on the two sides of the duality of discrete and continuous [3, Appendix A (The Duality of Discrete and Continuous)]: occupation counts in shares and is countable, the phase takes continuous values, and the two halves of the polar decomposition are exactly the two halves of the duality. The continuous coordinates of position belong to the readout (§1.4), and the values of the fields belong to the state space; a field takes its region-by-region form in the state space and appears in observation as a continuous field quantity. Motion is a process of the standing wave in the state space; a field is a state of the background in the state space. There are no fields in observational spacetime, only their appearance: the position-by-position values are read out through event statistics, and the limit of the statistics is the continuous field quantity of observation.

Fields and forces are two faces of one and the same coupling structure: a force is the mode of coupling between background and excitation, and a field is the position-by-position values of the background quantity through which the coupling passes. The forces of reference [1] are defined on the uniform vacuum, the metric force acting on occupation and the resonance forces on channel directions, with coupling strengths the same everywhere and no notion of position needed; matter makes the background quantities take values region by region, and the action of a force at each position is given by the value of the field there (§2.1, §3.2). Force in observation accordingly stands on no separate footing from the field: the deflection of a probe reads out the product of field quantity and charge (§1.1, §3.2), and the bending of a sequence is the single-probe reading of a field.

1.7 The Types of Fields

Fields fall into two types by background quantity: the spectral weight field $W(x)$, carried on the modular part of the polar decomposition, and the channel phase fields $a_k(x)$, carried on the involution part. The differences between the two types are four, each given by the algebra of one factor of the polar decomposition; if any one of them could not be so attributed, the classification would be falsified.

First, the sign is given by the two halves of the polar decomposition. Occupation is a non-negative count, the positive semidefiniteness of the modular part $\Delta^{1/2}$ agrees with it, and the deviation of the spectral weight has only the one direction of deduction,

$$n_{\text{occ}}(x) \;\geq\; 0 \quad\Longrightarrow\quad W(x) - N_{\text{vac}} \;=\; -\,n_{\text{occ}}(x) \;\leq\; 0\,, \tag{16}$$

and the interaction it carries has a single direction (attraction, §2). The charges of the channels come in two signs: the two members of a conjugate pair $\{\chi, \bar{\chi}\}$ are the two winding directions of one and the same standing wave [1], the antilinear involution J exchanges them, and the sign is read out by the difference of the two projections, $P_\chi - P_{\bar{\chi}}$, a linear involution with eigenvalues $s = \pm 1$, each member sitting on one eigenbranch. The coupling uses the sign twice, once in the accumulation by the source and once in the readout by the probe, and the product

$$s_1 s_2 \;=\; \pm 1 \tag{17}$$

takes both values: the interactions carried by channels come with both directions; the real direction and the emergent direction are fixed by J, have a single member, and carry no two signs.

Second, the dimension is given by the kind of source. The source of the spectral weight field is energy, valued in a continuous spectrum; the potential is dimensionless while the source density carries the fourth power of energy, so the coupling must carry energy to the power -2. The source of a channel phase field is charge, counted in shares with its sign taken from the two members of the conjugate pair, a pure number, so the coupling is dimensionless:

$$[G] \;=\; \text{GeV}^{-2}\,, \qquad [\alpha] \;=\; 1\,, \tag{18}$$

the divide of dimensions is the divide between continuous spectrum and discrete group.

Third, linearity is given by whether the carrier carries its own charge. The right-hand side of a source equation has two parts, the external charge and the contribution of the carrier itself,

$$\nabla^2 a \;=\; -\,J_{\text{ext}} \;-\; q_{\text{carrier}}\,\rho[a]\,, \tag{19}$$

where q_{carrier} is the charge of its own channel carried by the carrier and $\rho[a]$ the density of carrier excitations of that channel; the self-sourcing term exists if and only if $q_{\text{carrier}} \neq 0$. The color carrier carries the color label [1], $q_{\text{carrier}} \neq 0$, and the equation is nonlinear; the electromagnetic carrier is neutral, $q_{\text{carrier}} = 0$, and the equation is strictly linear; the transition quantum of the nuclear force propagates through the χ_0 channel, χ_0 is the trivial label, $q_{\text{carrier}} = 0$, and the equation is linear; the weak-channel equation is of linear Yukawa form, the question of carrier self-coupling being confined outside the readout layer by the force range (§3.4); gravity has no energy-carrying carrier (§2.6), the self-sourcing term has no bearer, the potential equation is strictly linear, and nonlinearity enters only through the relation between variable and readout (§2.5).

Fourth, the quantization of radiation is given by the presence or absence of a carrier. Channel radiation is the emission of carrier excitations, countable share by share; the adjustment of the metric has no corresponding excitation and travels outward as a continuous change of the readout. The difference is attributed in §2.6 and §3.7 as propagation is established.

The channel phase fields divide by resonance channel. The channel identification of reference [1] is: the cross-factor channel is electromagnetism, the $\mathbb{Z}/2\mathbb{Z}$ factor is the weak interaction, the $\mathbb{Z}/3\mathbb{Z}$ factor is color, and the collective coupling of the χ_0 emergent direction is the nuclear force; there are therefore four channel phase fields, and together with the spectral weight field each of the five interactions has its own field. The grounds of the identification, the charges, and the conservation laws are established in §3.1.

The classification is collected field by field as:

Field	Type	Channel	Source	Linearity	Section
Gravity	spectral weight ($\Delta^{1/2}$)	χ_0 ensemble sharing	energy	strictly linear	§2
Electromagnetic	channel phase (J)	cross-factor	electric charge	linear	§3.3
Weak	channel phase (J)	$\mathbb{Z}/2\mathbb{Z}$ factor	weak charge	linear	§3.4
Color	channel phase (J)	$\mathbb{Z}/3\mathbb{Z}$ factor	color charge	nonlinear	§3.5
Nuclear	channel phase (J)	χ_0 emergent	nucleon number	linear	§3.6

The predictions are collected in §6. The entire input to what follows is the two structures already established, the locality of the corrections and the region-by-region balance (§1.3), together with the numbers already fixed in reference [1]: the full-group energy scale $E_{\rm Pl} = 1.2215 \times 10^{19}$ GeV and the coupling constants G and α.

2 The Gravitational Field

The gravitational field is the spectral weight field, the position-by-position distribution in the state space of the background quantity of the metric structure. The standing waves of matter occupy the spectral weight of the region they sit in, the occupation lowers the cycle rate of the region below the vacuum value, and the time calibration of all excitations slows accordingly. There is no gravitational field in observational spacetime, only the statistics of events caused by gravity: redshifted spectral lines, bent light rays, falling trajectories, and the limit of their statistics is the reading of the potential and the metric. The equation of the distribution, the strong-field form, and the propagation in time are the content of this chapter.

2.1 The Gravitational Potential

The gravitational potential is given by the relation between occupation and energy. Occupation takes place on the metric structure. The metric force acts on all excitations, depends only on energy, and responds on a static background with an equilibrium distribution [1]; the equations of its dynamics that interface with this chapter are

Equation	Form	Content
Staticity	$\sigma_t(F_{\rm grav}) = F_{\rm grav}$	static background
Universality	$\partial F_{\rm grav}/\partial k = 0$	independent of character direction
Equilibrium distribution	$w_\beta(n) = e^{-\beta H(n)}/Z(\beta)$	response as a distribution, not a trajectory
Poisson identity	$-\Delta(\log n) = \log \frac{n^2}{n^2-1}$	intrinsic structure of the vacuum metric

Universality makes the occupied shares depend only on the energy of the standing wave, written $n_{\rm occ} = f(E)$ with E the energy of the standing wave; staticity, the equilibrium distribution, and the Poisson identity are used in the steady-state constructions of §2.2 and §2.5.

The form of f is fixed by additivity, its objects being bound sources in a common rest readout. Two widely separated standing waves each occupy shares in their own regions, and the shares add side by side; a composite standing wave within one region propagates as a whole in single cycles, with exchange phase M/v per cycle, M including the binding shares [3], and occupation adds with M. Hence

$$f(E_1 + E_2) \;=\; f(E_1) + f(E_2), \tag{20}$$

f is non-negative and non-decreasing in energy, the monotone solution of the Cauchy equation is linear, $f(E) = \lambda E$; summing the occupation of the standing waves in a region,

$$n_{\mathrm{src}}(x) \;=\; \lambda\, \rho_E(x)\, \delta^3, \tag{21}$$

with ρ_E the energy density and δ^3 the region volume; the proportionality constant λ is absorbed together with the coupling normalization in §2.3. n_{src} is the share matter occupies directly in the region, nonzero only at the source; in the steady state the local deficit $n_{\mathrm{occ}}(x)$ of the background is pinned at the source by n_{src} and extends beyond the source by redistribution (§2.2), and it is the latter that enters the deduction in (22). If occupation were counted by the squared-fluctuation convention, the strength of a composite source would change with its composition, contradicting the tests of the equivalence principle; the exclusion of the squared form is given in Appendix A; unbound free streams are not in this class, and their occupation is given separately by the measure of §4.2.

The measure of occupation is fixed by the classification of cycle phase. The phase of a cycle falls into two classes: the translation share is exact and costs nothing, while the exchange share is produced by spectral evaluation of the convolution and consumes carrying [3]; the number of consumed shares is proportional to the amount of exchange phase, the exchange phase of a composite per cycle is M/v, and occupation is proportional to M, which is exactly (21). The exchange share of a single free standing wave is $m^2/(vE)$, so the consumption rate is proportional to m^2/E, diluted in motion by the same factor as the slowing of proper time; the full derivation of the measure and the exclusion of the counting candidates are given in §4.1.

Occupation determines the local cycle rate. Substituting the effective spectral weight of the region (3) into the matching condition (1),

$$v(x) \;=\; E\big(N_{\mathrm{vac}} - n_{\mathrm{occ}}(x)\big), \tag{22}$$

where the measure of each share of fluctuation in the energy-scale function takes its vacuum value (§1.2); $E(W)$ increases monotonically with W, so occupation lowers the local cycle rate. The gravitational potential is defined as the relative deviation of the rate,

$$\Phi(x) \;=\; \frac{v(x) - v_0}{v_0} \;\leq\; 0, \tag{23}$$

$\Phi < 0$ where matter sits and in its surroundings. The cycle rate is the time calibration: each cycle registers one step, the step is $1/v$, the same for all excitations, and proper time is the quotient of phase count by mass [3]. A lowered $v(x)$ means a longer proper-time step, the time calibration near matter slows, which is gravitational redshift, with the sign agreeing with observation [27]. In the weak field $n_{\mathrm{occ}} \ll N_{\mathrm{vac}}$, Φ is proportional to n_{occ}, and the linearity of the source (21) passes to the potential; the distribution of the potential over position is given by the redistribution of the next subsection.

2.2 The Source Equation

Occupation falls in the region where matter sits, while the potential extends to distant places; the redistribution between the two gives the source equation.

Redistribution proceeds through the condensation direction of the background. The participating range of the emergent direction χ_0 is all sites of the full group, from χ_0 recoupling to every sector is possible, and redistribution acts only as an additive, sector-by-sector shift of weights [1]. The spatial reading is the same: the occupied shares of one region are transferred through the χ_0

direction to neighboring regions, the transfer is an additive shift, and shares are neither created nor destroyed. The three origins of conservation are: the total of all fluctuations C^* is invariant under time evolution [1]; shares are defined region by region (the diagonality of (8)); transfer is constrained by the propagation bound (6), with no direct transfer across a distance. The three give the continuity equation of shares

$$\partial_t n_{\mathrm{occ}} + \nabla\cdot\vec{J} \;=\; s, \tag{24}$$

$\vec{J}$ being the transfer flow of carrying and s the consumption rate of carrying (proportional to n_{src} and the cycle rate). The standing waves of matter consume one share of the background's carrying per cycle, and the carrying is transferred site by site along the network from afar: the steady state is a stationary regime (constant flux, static distribution), compatible with staticity, which excludes time variation but admits a steady flow. The deficit distribution $n_{\mathrm{occ}}(x)$ is maintained by this transport, its shape determined by the constitutive relation of the transfer (§2.5), isomorphic to the stationary temperature field of a heat-conducting rod held at fixed temperatures at both ends.

The shape of the steady distribution is fixed by the dispersion of propagation. The dispersion relation of cyclic propagation is $\cos(E/v) = \cos(p/v)\cos(m/v)$ [3]; the static field is its zero-frequency section: setting $E = 0$, the spatial wavenumber is purely imaginary, $p = i\kappa$, with $\cosh(\kappa/v)\cos(m/v) = 1$ and decay rate $\kappa = v\,\mathrm{arcosh}[\sec(m/v)]$, which for $m/v \ll 1$ is $\kappa = m[1 + O((m/v)^2)]$; the readout-level equation takes the leading form, and the static distribution decays spatially at the rate set by the carrier mass. The section argument rests on the propagation structure of the channels, whose microscopic form is given in §2.6; the gravitational potential is a stock distribution of the weight layer, whose nonlinear level is given separately by the constitutive relation of the transfer (§2.5), and the two routes coincide in the weak field as the Laplace equation. In the continuous coordinates of the readout, the static field quantity u outside the source satisfies

$$\left(\nabla^2 - m_{\mathrm{carrier}}^2\right) u \;=\; 0, \tag{25}$$

the shape of the equation depending only on the mass of the direction that carries the propagation. The redistribution of shares goes through the χ_0 direction, which matches automatically at zero frequency [1], so the decay rate is zero and the steady share distribution satisfies the Laplace equation. The distribution of a point source is uniquely fixed by conservation: the carrying flux through any enclosing surface equals the consumption rate inside, proportional to the total energy of the source, divided equally over the sphere in the three-dimensional readout, so the shape of the potential is $1/r$. Writing M for the total energy of the source,

$$\oint \nabla\Phi\cdot d\vec{A} \;=\; 4\pi G\, M, \tag{26}$$

whose differential form is the Poisson-type source equation [18]

$$\boxed{\nabla^2\Phi \;=\; 4\pi G\,\rho_E} \tag{27}$$

with the proportionality constant G fixed in the next subsection. The discrete Poisson identity of reference [1] gives the intrinsic structure of the vacuum metric, in which the source density is a derived quantity; (27) is its matter version, with the source as genuine input, the two adding layer by layer without conflict; the exact form of (27) in the state space is the region-by-region difference structure of the same type (§1.4).

The source variable is the energy density and no other combination. Occupation depends only on energy (§2.1), so pressure (the reading of momentum flux) does not enter the source; for comparison, the static effective source of general relativity [20] is $\rho + 3p$. The two are fully equivalent outside bound bodies: inside a bound body the pressure terms and the binding energy cancel in the total, only the total energy M is read outside, and solar-system tests do not distinguish them. The microscopic mechanism of the cancellation (the constraining interactions belong to the exchange share and make up the difference between the components and the total) is given in §4.2.

Unbound free streams have no such compensation, and their occupation density is the trace of the energy-momentum tensor, $\rho - 3p$ (derived in §4.2); the exchange share of light is zero, and freely propagating light sources nothing while responding fully (§4.3). The fork for free streams (the source is the trace $\rho - 3p$ here, $\rho + 3p$ in general relativity) and the interior structure equation of compact stars are both discriminable but as yet unjudged quantities (§6).

Pressure still enters the balance of forces: momentum flux takes part in the equation of motion, given by kinematic inheritance [3]; it merely does not source. What it does not enter is the source of the static potential equation; the source tensor of the dynamical wave equation contains energy flux and momentum flux by the Lorentz readout (§2.6), each living on its own level, with no conflict at leading order.

2.3 Newton's Constant

The proportionality constant of the source equation is Newton's constant. Φ is dimensionless and the coordinate carries the dimension of inverse energy, so the dimension of G is energy to the power -2; which energy scale fixes it depends on which structure carries the source equation. Occupation and redistribution proceed entirely on the metric structure: occupation depends only on energy (§2.1), and redistribution goes through the χ_0 direction, acting on all modes. A process carried by the metric has the full group as its participating set, with spectral weight $N = p^K$; for a system of spectral weight N the energy-scale formula gives $E(N)$, and the metric-force constant is defined from it [1]:

$$G \;=\; \frac{1}{E(N_{\mathrm{Pl}})^2} \;=\; \frac{C^*}{N_{\mathrm{Pl}} \log^2 N_{\mathrm{Pl}}} \;=\; 6.70 \times 10^{-39}\ \mathrm{GeV}^{-2}, \tag{28}$$

where $N_{\mathrm{Pl}} = 7^{41} \times 1.0271$ is the effective spectral weight of the full group (the value of p^K at $p = 7$, $K = 41$, times the Maschke-defect correction of reference [1]) and $E(N_{\mathrm{Pl}}) = E_{\mathrm{Pl}}$. Both the definition and the value are taken as given in reference [1]; the proportionality λ of (21) and the normalization of the share flow are absorbed here at once, the only observable combination being GM/r, with no intermediate quantity appearing on its own.

The division of labor between the participating sets is then clear: the absolute value of v_0 is given by the deduction on the unit group ((2)), and the scale of Φ is given by the energy scale of the full group ((28)), that is, the two participating sets of the energy-scale formula [1] in their places within the source equation. The potential of a point source is then fully determined,

$$\boxed{\Phi(r) \;=\; -\frac{GM}{r},} \tag{29}$$

the Newton potential [16], with G Newton's constant.

2.4 The Metric Readout

The coordinate time t is the time of the distant observer, and all rates, fluxes, and conservation statements are counted in t; the proper-time step at position x is $1/v(x)$, with $d\tau/dt = v(x)/v_0$. The spatial coordinate interval is the vacuum spacing $1/v_0$, and the proper length of one site at x is $1/v(x)$. Energies are always the readings at a distance: the energy of a standing wave at rest at x is $m(x) = m\, v(x)/v_0$, its reading at the reference state [1]; by local standards the mass of the same standing wave is the same as in the vacuum, $m(x)/v(x) = m/v_0$. The potential energy $m(x) - m = m\Phi$ is therefore counted once, and a probe moves along the stationary-phase line with its fixed local mass [3].

The observational content of the potential is read out through two rulers, and the two rulers compose the metric. The time ruler was given in §2.1: the proper-time step is $1/v(x)$, its ratio to that at infinity is $v_0/v(x)$, and the time component of the metric is

$$g_{00} \;=\; \Big(\frac{v(x)}{v_0}\Big)^2 \;=\; (1+\Phi)^2. \tag{30}$$

The spatial ruler is defined by the propagation of light: the spectral spacing of the background is not altered by matter (the metric structure is not determined by the distribution of excitations [1]), light travels one site per cycle [3], the proper length of one site is $1/v(x)$, while the coordinate interval is the vacuum spacing, so the spatial component is locked by the time component, $g_{rr} = (v_0/v(x))^2 = 1/g_{00}$, and in the weak field $g_{rr} = 1 - 2\Phi$.

The ruler of matter gives the same reading: all masses of the mass spectrum are v times universal number factors [1], after localization $m(x) \propto v(x)$, and the sizes of all bound structures (atomic radius $\propto 1/(m\alpha)$, nuclear size $\propto 1/m_\pi$) scale as $1/v(x)$. Any ruler built from the local mass spectrum gives the same spatial component as light calibration; the locking is the consistency of the readout. The position dependence of mass at once gives the attribution of potential energy: the mass deficit of a bound system, $m(x) - m(\infty) = m\Phi = -GMm/r$, is the Newtonian potential energy; potential energy belongs to the local mass of matter, and the region of the field carries no energy. The weak-field metric assembles as

$$\boxed{ds^2 \;=\; (1+2\Phi)\,dt^2 \;-\; (1-2\Phi)\,d\vec{l}^{\,2},} \tag{31}$$

time and space components equal and opposite, two readouts of one and the same potential; the full equation of the spatial component is established independently in §2.5.

The response is inherited in full from reference [3]: the transport equations give the stationary-phase line of a standing wave in a region where $v(x)$, $\delta(x)$ vary slowly, position dependence enters all excitations through the same factor, all standing waves fall alike, and the equivalence principle holds; slow excitations register only the position dependence of the cycle rate, light registers both, and the deflection of light is twice the bending of slow trajectories [24]. The $2:1$ ratio is given independently in reference [3] from phase shares, consistent with the consequence of the metric (31); the two routes meet here. Stationary phase is wave-packet motion, and in the weak field it agrees with the geodesics of the metric (31) (the eikonal correspondence),

$$\frac{d^2x^\mu}{d\tau^2} + \Gamma^\mu_{\alpha\beta}\, u^\alpha u^\beta \;=\; 0\,, \tag{32}$$

Γ being the Levi-Civita connection of (31); the equation is the weak-field form of stationary phase. The parametrized readings of solar-system tests are thereby legitimate: (31) gives $\gamma = 1$, and together with the second-order expansion of §2.5 gives $\beta = 1$, consistent with all existing observations [9].

2.5 The Nonlinear Equation and the Closed Form

The weak-field results receive corrections near the source, and the corrections originate in redistribution itself taking place on the background being corrected: the cycles the transfer passes through are the local cycles, so the coefficients of the equation change with the field. In the static isotropic problem the spatial component is already locked by the time component (§2.4), and the only dynamical field is $v(x)$; the full equation and its output are established in four steps, each using only structures already in place.

Step 1: the shape of the equation is given by the equilibrium measure of the exchange network. The steady distribution is the stationary regime of carrying transport (§2.2), and the transport runs on the network of events: the positions of shares are carried by event readings (§1.3), the nodes of the walk are event slots, a transfer is a jump between neighboring event slots, and all rates are counted in the common coordinate time (§2.4). The physical input of the network is two items. One is detailed balance: each exchange is a single cyclic process of outflow and retrieval [1], the symmetry of the two-way rates is given by the two-half structure of the exchange, and the jump rates satisfy $w_{ij}\pi_i = w_{ji}\pi_j$, with π the source-free stationary measure. The other is the uniform walk of shares on the event network: the node density is the number of events per site per unit coordinate time, the proper cycle rate v times $d\tau/dt = v/v_0$, proportional to g_{00}; each event slot carries equal weight, the spatial form of equal sharing [1], so $\pi \propto g_{00}$; the jump rate between neighboring event slots is uniform in coordinate time, so the conductance $\kappa_{ij} = w_{ij}\pi_i$ is proportional to g_{00}. The stationary state satisfies exactly, on the lattice,

$$\sum_j \kappa_{ij}\Big[(n/\pi)_j - (n/\pi)_i\Big] \;=\; 0\,, \tag{33}$$

whose differential form for statistical field quantities is

$$\nabla\cdot\big[\kappa\nabla(n/\pi)\big] \;=\; 0\,, \tag{34}$$

with a source its solutions carry a constant flux. Substituting the channel count of the deficit, $n \propto 1 - W/N_{\rm vac} = 1 - g_{00}$ (the measure convention gives $W/N_{\rm vac} = g_{00}$, §1.2), the spherically symmetric stationarity condition is

$$\kappa\Big(\frac{n}{\pi}\Big)'r^2 \;\propto\; g_{00}\Big(\frac{1-g_{00}}{g_{00}}\Big)'r^2 \;=\; -\,\frac{g_{00}'\,r^2}{g_{00}} \;=\; \text{const}\,, \tag{35}$$

with equation and solution

$$\frac{d}{dr}\,\log g_{00} \;\propto\; \frac{1}{r^2} \qquad\Longrightarrow\qquad \boxed{\;g_{00} \;=\; e^{-2GM/r}, \qquad g_{rr} \;=\; e^{2GM/r}\,,\;} \tag{36}$$

that is, the logarithmic potential $\log\big(v(x)/v_0\big) = -GM/r$ is harmonic, and the weak-field truncation is the source-free form of (27). The logarithm is the intrinsic variable of the background metric ($H = \log n$, distance $d(m,n) = |\log(m/n)|$, everything counted in logarithms [1]): in the logarithmic variable the equation returns to the strict Laplace form, the potential carries no source of its own, and matter is the only source. The timing convention is selected by observation: if the jump rate were uniform in the local proper time, converting to coordinate time gives $\kappa \propto g_{00}^{3/2}$, with solution $g_{00} = (1-GM/r)^2$ and second-order coefficient $+1$, contradicting the perihelion precession (Appendix A).

Step 2: observation excludes the alternative measures listed. Parametrizing the coefficients of the network as $\kappa \propto g_{00}^a$, $\pi \propto g_{00}^b$, the spherically symmetric stationarity condition of (34) gives $g_{00} = 1 - 2x - 2(a-2b)x^2 + O(x^3)$ ($x = GM/r$, Appendix A), and the perihelion precession requires the second-order coefficient $+2$, that is, $a = 2b-1$. The uniform walk $(1,1)$ lies on the line; independent per-share jump rates $(0,-1)$, interface-fixed rates $(1,0)$, and complete mixing $(2,1)$ give -4, -2, 0, the jump rate uniform in proper time $(3/2,1)$ gives $+1$, and the three flat formulations that write the transfer on a rigid coordinate lattice are all negative, all excluded (Appendix A). The line holds more than one point: $(3,2)$ also passes the second order, with third-order coefficient -4, and $(-1,0)$ and $(1,1)$ give one and the same equation at all orders. The exponential form is therefore the solution of the uniform-walk network, compatible with existing observation; the third-order coefficient $-4/3$ is the falsifiable prediction of this measure.

Step 3: the order-by-order expansion completes the comparison with general relativity, carried out in $x = GM/r$ (both in isotropic-type coordinates, asymptotically normalized, GM defined by the first order; the Schwarzschild solution [21] taken in isotropic coordinates):

	this paper g_{00}	GR g_{00}	this paper g_{rr}	GR g_{rr}
x	-2	-2	$+2$	$+2$
x^2	$+2$	$+2$	$+2$	$+3/2$
x^3	$-4/3$	$-3/2$	$+4/3$	$+1/2$

$\gamma = 1$ (light deflection and time delay) and $\beta = 1$ (perihelion precession) are consequences, and all existing solar-system tests pass [9]; the deviation from general relativity starts at the second order of the spatial component ($+2$ versus $3/2$) and the third order of the time component ($-4/3$ versus $-3/2$), originating in the absence of the self-sourcing structure of the field (§6). The order-by-order coefficients depend on the coordinate gauge convention (the table is in isotropic-type gauge), and the comparison of observables removes this dependence: the orbit of light depends only on the ratio g_{rr}/g_{00}, which under the locking is $1/g_{00}^2 = e^{4GM/r} = 1 + 4x + 8x^2 + \cdots$, and the deflection angle in terms of the invariant impact parameter b is

$$\alpha \;=\; 4\,\frac{GM}{b} \;+\; 4\pi\Big(\frac{GM}{b}\Big)^2 \;+\; O\Big(\Big(\frac{GM}{b}\Big)^3\Big), \tag{37}$$

the second-order coefficient of general relativity being $15\pi/4$: the relative deviation is $+1/15$, at solar grazing 11.7 versus 10.9 μas, discriminable by next-generation astrometry (derivation and numbers in Appendix B).

Step 4: the strong-field conclusions are read off the closed form. Occupation is a count, non-negative and unable to exceed the total spectral weight, so the potential has the lower bound $\Phi \geq -1$; the solution (36) satisfies it everywhere, and $g_{00} = e^{-2GM/r}$ is positive for all $r > 0$: there is no horizon at finite radius, and the redshift grows without bound as r decreases. The domain of validity of the closed form ends where the continuous approximation holds (§1.4): the scale of variation of the field, $r^2/2GM$, shrinks as r decreases, steady-state localization fails where it reaches the order of the local spacing $e^{GM/r}/v_0$, and the structure of the deep region is not given by the closed form, left to later work. The observable at horizon scale is the critical impact parameter of light: the orbit equation of light gives $b_c = \min_u F(u)/u$, the extremum of $F = e^{2GMu}$ is at $u^* = 1/2GM$,

$$b_c \;=\; 2e\,GM \;=\; 5.437\,GM, \tag{38}$$

general relativity gives $3\sqrt{3}\,GM = 5.196\,GM$: the shadow diameter is larger by 4.6%, compatible with current horizon-scale imaging (§6), discriminable by the next generation; the formation of the shadow further requires the optical conditions of the interior, and this paper gives no interior solution.

2.6 Waves and Radiation

Waves are the propagation in time of the adjustments of the field. When the source varies in time, the adjustment of $v(x)$, together with the metric readout it locks, propagates outward. The level of the propagation decides the type of the equation: the adjustment proceeds through the cyclic exchange of the background, which is amplitude-level [3], the generator is unitary, the spectrum purely imaginary, and it propagates as traveling waves at one site per cycle, the speed of light; probability-level readouts (the relaxation of occupation) are driven by dissipative generators [2], with real negative spectrum, decaying without propagating. There are only these two levels: superposition splits into by-amplitude and by-intensity, intermediate cases are excluded by the orthogonality of conservation sectors and the modular-flow average [1], and each component falls into one of the two according to its readout. The dynamical content comes in four items, each given by this level structure together with the conservation laws.

The freely propagating components are identified by the structure of the readout. In the three-dimensional readout, light calibration proceeds direction by direction, the spatial part of the metric is a symmetric g_{ij}, and the static isotropic case returns to $g_{rr}\,\delta_{ij}$; writing the perturbation as h_{ij} and decomposing with respect to the propagation direction $\hat{z}$ into six components,

$$\underbrace{h_{zz},\ h_{xz},\ h_{yz}}_{\text{longitudinal}}\,; \qquad \underbrace{h_{xx}+h_{yy}}_{\text{transverse trace}}\,; \qquad \underbrace{h_{xx}-h_{yy},\ h_{xy}}_{\text{transverse traceless}}\,. \tag{39}$$

The three longitudinal components are locked to the source: the rearrangement of spacing along the propagation direction is constrained by flux conservation, the carrying flux through each cross-section is fixed by the sources inside (the dynamical version of (26), $\nabla\cdot\vec{J} = -\,\partial_t n_{\text{occ}}$), they carry no free wave, follow the variation of the source through the retarded structure at the same speed, in the same structure as the locking of the longitudinal components of the electromagnetic field to the charge; the dynamical adjustment of the potential belongs to these slaved components, and the probability-level relaxation criterion applies only to free modes. The transverse trace component belongs to the weight level and decays without traveling outward: the trace is the density readout of carrying shares, a probability-level conserved count, its evolution driven by a dissipative generator with real negative spectrum ($\partial_t u_{\text{tr}} = -\Gamma\, u_{\text{tr}}$, $\Gamma > 0$); the separate relaxation of occupation-type and coherence-type readouts in reference [2] provides the precedent of the same type. The two transverse traceless components are the only free radiation modes: the relative phase structure between directions, amplitude-level, propagating unitarily. The radiation field therefore contains only the two transverse traceless polarization components, consistent with the observation of gravitational-wave polarizations [13]; the propagation speed is the speed of light, consistent with the bound on the speed difference given by the near-simultaneous arrival (1.7 s apart) of the gravitational wave and the gamma-ray burst of the binary neutron star merger [14].

The propagation equation of the free components is fixed by the microscopic structure. Propagation at the amplitude level is the alternation of two branches: the left-moving and right-moving components each shift one site per cycle, formation exchange mixes the two branches, the mixing angle is the phase per cycle, and the dispersion relation $\cos(E/v) = \cos(p/v)\cos(m/v)$ is exactly the spectrum of this structure [3]. The adjustment of the metric is carried on the χ_0 direction, which matches at zero frequency [1], so the mixing angle is zero and the dispersion degenerates,

$$\cos(E/v) \;=\; \cos(p/v) \quad\Longrightarrow\quad E = \pm p\,, \qquad \frac{dE}{dp} = \pm 1\,, \tag{40}$$

the two branches decouple into pure translation, the equality holds on the lattice, and the zero-mass channel has no discrete dispersion correction. The evolution is unitary, the equation linear in the amplitude, and outside the source, in the readout on a uniform background with constant v,

$$\left(\partial_t^2 - \nabla^2\right) h_{ij}^{TT} \;=\; 0, \tag{41}$$

the coefficient of the source coupling locked by the static limit (§2.2). Three consequences are read off directly: gravitational waves and light share one and the same propagation structure, both pure translation of the zero-mass channel, their speed difference exactly zero, so the 10^{-15} bound on the speed difference from the binary neutron star merger [14] tests an identity; on a slowly varying background the step and the step rate of the walk are both local quantities, so gravitational waves refract in a potential well exactly as light does; the linearity of the equation in the amplitude is a consequence of unitarity, nonlinearity enters only through the background $v(x)$, of the same origin as the static conclusion. One and the same exchange network thus carries two levels of dynamics, a unitary walk on the amplitude level and a dissipative master equation on the weight level [2]: the separate relaxation of occupation-type and coherence-type readouts reappears in the spatial readout, and the non-radiating trace and the radiating transverse traceless components come from the two levels of one microscopic model.

A moving source leaves mixed components in the metric. When the source moves as a whole at slow velocity w_i, the readout of the static solution acquires g_{0i} under the Lorentz transformation [1]: the time and space components are equal and opposite ((31)), the two contributions add with the same sign, and with the convention that ds^2 contains $2g_{0i}\,dt\,dx^i$,

$$g_{0i} \;=\; -4\,\Phi\,w_i, \tag{42}$$

$\Phi < 0$ making the component parallel to the source velocity. The coefficient 4 is the direct mark of the spatial component taking part in the readout: the same operation in electromagnetism has only the time component, with coefficient 1. A rotating source superposes volume element by volume element (linearity of the source, §2.1), giving frame dragging in agreement with general relativity; the gyroscope precession measurement [8] and satellite laser ranging pass here with zero parameters, providing a second test of the spatial-component readout independent of light deflection.

The multipoles and the power of radiation are given by the conservation laws and a source-side derivation. Conservation of total energy excludes monopole radiation (the monopole moment $\int \rho_E = M$, $\dot{M} = 0$); conservation of momentum [3] makes the center of mass move uniformly and excludes dipole radiation (the second time derivative of the dipole moment $\int \rho_E\, x^i$ is $\dot{P}^i = 0$); the lowest radiating moment is the quadrupole. The quadrupole power is derived directly on the source side: rather than first defining the energy density of the wave, one computes the energy lost by the source. The component structure of the wave equation is fixed by the static limit and the Lorentz readout: the source coefficient $-16\pi G$ is locked by the Poisson limit, energy density, energy flux, and momentum flux compose the source tensor (assembled by the same method as §3.3), and the difference of pressure not sourcing does not appear at leading order. The expansion of the retarded solution in the source region is pure mathematics ($\bar{h}_{ij}$ the components of the wave equation, T_{ij} the corresponding components of the source tensor, $R = |\vec{x} - \vec{x}'|$),

$$\bar{h}_{ij}(\vec{x}, t) \;=\; 4G\int \frac{T_{ij}\big(\vec{x}',\, t - R\big)}{R}\, d^3x' \;=\; 4G\sum_{k\geq 0} \frac{(-1)^k}{k!}\,\partial_t^k \int T_{ij}\, R^{\,k-1}\, d^3x'\,, \tag{43}$$

the even orders give near-zone corrections and the odd orders the radiation-reaction potential, with quadrupole leading term

$$\Phi_{\rm react} \;=\; c\,G\,x^i x^j\,\frac{d^5 Q_{ij}}{dt^5}, \tag{44}$$

Q_{ij} the traceless quadrupole moment and c a pure number: the $k = 5$ traceless part of the expansion gives this form, the value of c requires the retarded structure of all slaved components to be counted together, which this paper does not give, and the same computation in the linearized theory gives $c = 1/5$, for comparison. The reaction force is the negative gradient of this potential (the same expression as $\vec{F} = -q\nabla a$ of §3.2), and the average of its work on the source (integrating by parts twice, boundary terms vanishing under the time average) gives

$$P = cG \langle \dddot{Q}_{ij} \dddot{Q}_{ij} \rangle, \tag{45}$$

carried out on the source side by mechanics; once the coefficient is fixed, the observed orbital decay of the binary pulsar [7] is its test. The energy density of the radiation field is a derived quantity: energy conservation hands the power lost by the source to the radiation field, the conversion of the far-field solution gives the customary density expression, and its coefficient is the same constant as the source coefficient. The observed field quantities are statistical readings, energy conservation closes between source and matter, and the energy density of the field is a statistic derived from conservation; the difficulty that gravitational field energy can be localized nowhere does not arise.

The carrier and the energy of the wave are given by the established conclusions and the fluctuation weights. The excitations that can carry records pass the triple sieve of conservation, energy, and record, and under the particle identification of the same framework only the photon remains [2]; the metric structure has no excitation of the propagation operator [1]: in this framework gravitational waves are not carried by quanta; they are collective adjustments of the metric readout; the corresponding mode of observation is then self-consistent, a gravitational wave produces no events but changes the statistics of photon events, and what the interferometer reads is the latter. The energy of the wave is derived from conservation: the power lost by the source, (45), is handed to the radiation by energy conservation and received where it arrives by the event-by-event balance of the detector [3]; the conversion of the far-field solution gives the expression for the energy density, scaling as the square of the amplitude, with dimensional structure $\dot{h}^2/G$ and the same constant as the source coefficient. The fluctuation weight itself is unchanged by a traceless modulation: the squared proper spacing in each direction is $\delta l^2(1 + h_i)$, the sum is $\delta l^2(3 + \sum_i h_i)$, and $\sum_i h_i = 0$ removes the first order and all higher orders at once; the wave does not change the fluctuation weight of the regions it passes through and sources nothing, of the same kind as the zero sourcing of light (§4.3), consistent with gravity having no energy-carrying carrier. The static potential and the free wave are of different microscopic origin: the potential is the stock distribution of steady transport on the weight level, the wave is propagation on the amplitude level; the applicability of the dispersion section (§2.2) to the potential is therefore confined to the weak-field level, the nonlinear level being borne by the constitutive relation alone.

3 The Electromagnetic, Weak, Nuclear, and Color Fields

The electromagnetic, weak, nuclear, and color fields are all channel phase fields, the position-by-position distributions of the phase of each channel. A standing wave carrying charge accumulates channel phase along its dwell segments, and the accumulation makes the phase of the region deviate from the vacuum zero; the distribution of the phase is read out only by probes carrying the same kind of charge, in contrast with gravity, which is read out jointly by all observations; in observational spacetime there are only the statistics of the events of charges, and the field strength is the reading of these statistics (§3.7). The structure of the charges, the equations of the field distributions, the ranges of the channels, and the readout in observational spacetime are the content of this chapter.

3.1 Channels and Charges

The charge of a field is labeled by its channel. The resonance channels divide by CRT factor: the unit group is $(\mathbb{Z}/7\mathbb{Z})^* \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$, the resonance forces act factor by factor independently, and there is one additional cross-factor channel [1]; the identification of reference [1] is: the cross-factor channel is electromagnetism ($U(1)$), the $\mathbb{Z}/2\mathbb{Z}$ factor is the weak interaction (carrier direction χ_3), and the $\mathbb{Z}/3\mathbb{Z}$ factor is color ($\chi_2 \oplus \chi_4$). The sign and the conservation law of the charge of each channel are given by the character label of that direction: the label is conserved through interactions, $\chi_{\text{in}} = \chi_{\text{out}}$, and the frequency label is the conserved quantum number of the interaction [1]; the magnitude of the charge counts in shares of standing waves of the same channel (§3.2). The response of a resonance force acts on the direction degrees of freedom, scaling the amplitude and deflecting the phase of the direction component while leaving the direction label itself unchanged [1]; in contrast with the universality of occupation, which depends on energy alone, the action of a channel selects by label.

The dynamics of the resonance forces on the state space is given by reference [1]; the equations that interface with this chapter are

Equation	Form	Content
Conservation law	$\chi_{\text{in}} = \chi_{\text{out}}$	conservation of quantum numbers
Matter response	$P_k K\psi = \lvert\lambda_k\rvert e^{i \arg \lambda_k} P_k\psi$	amplitude scaling and phase deflection, identity unchanged
Nonlinear resonance	$\partial_\beta A_k + \nu_k A_k + \sqrt{\alpha_k} \sum f_{ijk} A_i A_j = \mathcal{J}_k$	self-coupling and matter source
Joint coupling	$\alpha_{\text{mix}} = \lvert\text{Aff}\rvert_{\text{eff}}\, \lvert R_k\rvert^2$	emergent–elementary joint coupling

A_k is the resonance amplitude, ν_k the linear damping coefficient, $f_{ijk} = \delta_{i+j,k}$ the structure constants, $\mathcal{J}_k$ the matter source term, and $R_k = L(\beta, \chi_k)/\zeta(\beta)$ the resonance ratio. The three kinds of coupling obey the unified formula $\alpha_X = |A_X|^2 \eta_X$: the elementary resonance forces take $\eta = 1/N_k^{\text{eff}}$ (dilution by averaging), the emergent resonance force takes $\eta = |\text{Aff}|_{\text{eff}}$ (enhancement by summation) [1]; the resonance of the emergent direction is the nuclear force, whose field is confined to the nucleus together with the color channel (§3.6).

The two signs of charge come from the conjugate structure. The two members of a conjugate pair $\{\chi, \bar{\chi}\}$ are the two winding directions of one and the same standing wave [1], the positive and negative charges; they are each other's image under J, and the operator that reads out the sign is the difference of the two projections, $P_\chi - P_{\bar{\chi}}$, linear, an involution, with eigenvalues ± 1. The observable potential energy is the product of the accumulation by the source and the coupling of the probe, using the same sign twice: like charges give $(\pm 1)^2 = +1$, positive potential energy, repulsion; unlike charges give -1, attraction. The two-sign structure belongs to conjugate-pair channels; the real direction χ_3 and the emergent direction χ_0 are fixed by J, have a single member, and their signs are not given by the two-sign rule: the sign of χ_0 is given by its equation (attraction, §3.6), and the potential of χ_3 does not enter the macroscopic readout (§3.4), its sign having no readout counterpart. The metric has no involution structure and occupation is non-negative, so gravity is always attractive; the difference between like-charge repulsion and universal attraction is the difference between the algebras of the two parts of the polar decomposition, already attributed in §1.7.

3.2 The Phase Field

The phase field acts on charge-carrying matter through the phase per cycle: a standing wave of charge q at position x has a channel share $q\, a_k(x)$ in its phase per cycle (a_k is the channel phase of §1.2; this chapter takes the electromagnetic channel as the main case and drops the subscript k below). The principle of stationary phase acts on the total phase, and the total phase is the share-by-share sum of the metric share and the channel share [1], so the position-gradient force holds in the same form for every share of phase [3],

$$\vec{F} \;=\; -\,q\,\nabla a, \tag{46}$$

the electrostatic force; the full form felt by a moving charge follows with the readout of the field strength in §3.7. The accumulation of phase acts only on excitations carrying the charge of that channel; neutral excitations do not feel it.

The accumulation is linear in charge. The accumulation of phase proceeds standing wave by standing wave: each charged component of a composite accumulates along its own dwell segments, the insertion flows do not merge, the phase value of the region is the sum of the accumulations of the components, and the two winding directions accumulate with opposite signs. Counting charge in shares, the charge of a composite is the signed sum of the shares of its components, and the accumulation is linear in it:

$$a\big(q_1 + q_2\big) \;=\; a(q_1) + a(q_2). \tag{47}$$

The multiplication of labels $\chi_i \cdot \chi_j = \chi_{i+j}$ gives only the combinations allowed at a vertex and the conservation law (§3.1), not a many-body addition of charges: two shares of charge in the same channel accumulate two shares, not the sum of the labels. The linearity of gravity was derived from additivity through the Cauchy equation (§2.1); the linearity here is carried directly by the additivity of accumulation standing wave by standing wave, with no limiting argument needed; a squared form would leave a net source for neutral atoms, contradicting the tests of the electric neutrality of matter, excluded in Appendix A.

The absence of crosstalk between the two kinds of correction is guaranteed by their respective selectivities. The metric does not distinguish character directions [1], so charge does not enter the gravitational potential: a charged body gains no extra gravity by being charged; the insertion flow only accumulates phase without changing the modulus [1], so phase does not enter the cycle rate: the time calibration of a charged body is fixed by its energy alone. The energy that maintains the phase accumulation is itself counted into occupation by its energy, so the energy of the electromagnetic field takes part in gravity in the usual way; the two kinds of correction close each on its own variable.

3.3 The Electromagnetic Channel

The electromagnetic field is the phase field of the cross-factor channel, the main body of the long range; its static equation is the unified shape (25) evaluated at zero carrier mass. The carrier of the electromagnetic channel is the photon, $m_{\text{carrier}} = 0$ [1], so outside the source $\nabla^2 a = 0$. Charge conservation gives a flux conservation isomorphic to (26): the field flux through any enclosing surface equals the total charge inside, and the potential of a point charge is

$$\Phi_{\text{EM}}(r) \;=\; \frac{\alpha\, q_1 q_2}{r}, \tag{48}$$

the Coulomb potential [17].

The complete set of equations is assembled in three steps. Step 1: when the source moves, the charge density and the readout of the phase field mix into spatial components under the same Lorentz operation (the same method as the metric mixing of §2.6), and a_μ and the charge current J^μ form four-vectors. Step 2: the phase is observable only through interference between paths [3]: the phase $q\int a_\mu dx^\mu$ accumulated along the world line of a probe changes under $a_\mu \to a_\mu + \partial_\mu\theta$ only by endpoint terms, which belong to the phase convention of the probe itself and are unobservable; the equations can therefore contain only the antisymmetric gradient $F_{\mu\nu} = \partial_\mu a_\nu - \partial_\nu a_\mu$, and gauge invariance is the pointwise freedom of the phase zero (the vacuum value set to zero, §1.2); a_μ is the phase per cycle times the cycle rate and carries the dimension of energy. Step 3: charge conservation $\partial_\nu J^\nu = 0$ (label conservation [1]) forces the time-dependent terms: the naive extension of the static form violates conservation, and the completing terms are fixed uniquely by conservation. The unique equation in the variable F that is covariant, linear, of first order in the derivatives, and identically divergence-free on the left is

$$\boxed{\partial_\mu F^{\mu\nu} \;=\; J^\nu,} \tag{49}$$

whose left-hand side is identically divergence-free, consistent with charge conservation; the source-free set $\partial_{[\lambda} F_{\mu\nu]} = 0$ is an identity of the definition (Heaviside–Lorentz units, the charge in J^ν counted as $e = \sqrt{4\pi\alpha}$).

3.4 The Weak Channel

The equation of the weak channel is given by the carrier mass of the breaking direction. The breaking direction is uniquely χ_3, with carrier mass $M_W = v_0\sqrt{C^*/N_W^{\text{eff}}} = 80.6\,\text{GeV}$ [1]; the source is the distribution J_w of weak charge (the χ_3 label), with source equation

$$\left(\nabla^2 - M_W^2\right) a_w \;=\; -\,J_w, \tag{50}$$

the carrier carries the χ_3 label, so a self-coupling term exists; its effects are confined together with the field to subnuclear scales by the force range, the readout-level equation is the linear Yukawa leading form for static weak sources [26], and the pairing structure of the emitted pair belongs to the process side of decay [4]. The potential is of type $e^{-M_W r}/r$, with range

$$\frac{1}{M_W} \;\approx\; 2.4 \times 10^{-3}\ \text{fm}\,, \tag{51}$$

the short range given by the carrier mass, with no separate mechanism; the neutral component has carrier mass $M_Z = 91.4\,\text{GeV}$ [1], with range of the same order. At macroscopic distances the potential decays as $e^{-M_W r}$ to invisibility, and the weak-channel field does not enter the macroscopic readout.

3.5 The Color Channel

The equation of the color channel is nonlinear because the carrier carries its own charge. The breaking direction χ_3 does not touch the $\mathbb{Z}/3\mathbb{Z}$ factor, and the carrier stays massless [1]. The source is the distribution of color charge (the $\chi_2 \oplus \chi_4$ labels); the carrier carries the color label, the field couples to itself, and the source equation carries a nonlinear term,

$$\nabla^2 a_c \;=\; -\,J_c \;+\; (\text{self-coupling term}), \tag{52}$$

discriminably different from the linearity of electromagnetism (§1.7). The solutions are constrained by confinement: excitations are confined within color-neutral composites [1], the channel field does not extend beyond the color-neutral scale, and the source problem is confined to the nucleus. The state-space form of the self-coupling is the structure-constant term of the nonlinear resonance equation in the table of §3.1 [1], whose label multiplication is commutative; its readout-level form is not given by the readout of this paper, since confinement leaves no macroscopic sources or probes (§3.8).

3.6 The Nuclear Force

The equation of the nuclear force is given by the transition quantum of the emergent direction. Color-neutral composites (nucleons) no longer exchange color charge, and the remaining interaction proceeds through the emergent direction: the collective coupling of the χ_0 direction (§3.1) is carried between nucleons by the transition quantum, whose mass is $m_\pi = 138.3\,\mathrm{MeV}$ [1]. The potential field φ_N between nucleons satisfies

$$\left(\nabla^2 - m_\pi^2\right)\varphi_N \;=\; -\,g\,\rho_N\,, \tag{53}$$

ρ_N being the density distribution of nucleon number. The source counts in nucleon number, one share per nucleon: a nucleon is one share of collective condensation of the χ_0 direction [1], the transition quantum couples to the condensate through the χ_0 channel, and the number of shares is given by the count of the condensation. g is the nuclear coupling: $g^2/4\pi = 14.10$ takes the value of the unified formula for emergent resonance [1], the observed coupling of the nucleon–transition-quantum vertex; g is dimensionless, φ_N carries the dimension of energy, and the dimensions of the equation close by themselves. The potential is $e^{-m_\pi r}/r$, with range

$$\frac{1}{m_\pi} \;\approx\; 1.4\ \mathrm{fm}, \tag{54}$$

the measured range of the nuclear force. The transition quantum is a scalar ($J = 0$ [1]), and the leading form of scalar exchange gives the central potential $-g^2 e^{-m_\pi r}/4\pi r$; (53) is the leading form of single transition-quantum exchange, and the dependence of the nucleon potential on winding direction and line labels, together with its short-range part, lies beyond it. Reference [1] established the mass of the transition quantum but not the field equation of its propagation; (53) follows from the unified static equation (25) at $m = m_\pi$, its range of the same origin as the other channels. The χ_0 direction has two uses: the redistribution of the metric uses its ensemble sharing (§2.2), the zero-frequency weight relay giving the long range; the nuclear force uses its resonance excitation, propagating through the massive transition quantum and giving the short range. The two uses are distinguished by the carrier, and the ranges differ accordingly.

The static source equations of the five interactions are now all established, sharing one shape (the weak-field equation of gravity is the same shape in the zero-frequency case of the χ_0 direction; the exact nonlinear form is §2.5):

Channel	Carrier mass	Equation outside source	Potential
Gravity (metric structure, χ_0 ensemble relay)	0 (zero frequency)	log potential harmonic (§2.5)	$1/r$
Electromagnetic (cross-factor)	0	$\nabla^2 a = 0$	$1/r$
Weak ($\mathbb{Z}/2\mathbb{Z}$ factor)	$M_W = 80.6\,\mathrm{GeV}$	$(\nabla^2 - M_W^2)\,a = 0$	$e^{-M_W r}/r$
Nuclear (χ_0 channel, via transition quantum)	$m_\pi = 138.3\,\mathrm{MeV}$	$(\nabla^2 - m_\pi^2)\,\varphi_N = 0$	$e^{-m_\pi r}/r$
Color ($\mathbb{Z}/3\mathbb{Z}$ factor)	confined	with self-coupling, nonlinear	confined to nucleus

The fork lies solely in the mass of the direction that carries the propagation. The dynamical equation of the massive channels is the full form of the same dispersion,

$$\left(\partial_t^2 - \nabla^2 + m_{\text{carrier}}^2\right) a \;=\; J\,, \tag{55}$$

J being the source of the channel; the zero-frequency section is the static equation of the table above (§2.2), and at $m_{\text{carrier}} = 0$ it returns to the wave forms (49) and (41). The ranges then give the roster of fields entering the macroscopic readout: gravity and electromagnetism have massless carriers and are long-range; the weak and nuclear carriers are massive and short-range; color is confined to the nucleus by confinement. Everyday contact interactions all proceed through the electromagnetic channel, and the weak channel opens only in processes that change identity [3]. The only long-range fields in the macroscopic readout are gravity and electromagnetism, a roster consistent with observation and with the completeness of reference [1].

3.7 The Readout of Field Strength

The coupling constants are fixed by the same method as gravity: electromagnetism is carried by characters, the vertex response and the channel count give $\alpha^{-1} = 137.036$ [1], taken over directly here; the attribution of the difference in dimension was given in §1.7.

The readout forms a set of equations of its own. The field strength is defined by the events of probes: a probe of charge q reads out one balance of momentum at each record, the sums of readings before and after the scattering being equal [3], and the four-momentum increment the probe acquires between two events is attributed to the field. Writing Δp^μ for the increment, $\Delta\tau$ for the proper time between the two events, and u_ν for the four-velocity of the probe, the reading combination $f^{\mu\nu}$ of a single event is defined by $\Delta p^\mu = q\, f^{\mu\nu} u_\nu\, \Delta\tau$, its six components read out by probes of different velocities; the field strength is the statistical average over the region ensemble (§1.4),

$$F_{\mu\nu}(x) \;=\; \langle f_{\mu\nu} \rangle_{U(x)}\,, \tag{56}$$

which defines the field quantities of observation. The photon only carries the record: its energy, direction, and polarization give the position and momentum of the probe through the scattering [3], and the sign and phase of the field are read out from the deflection of the probe, not from the counting of the photons themselves. The component readout is

$$E_i \;=\; F_{0i}, \qquad B_i \;=\; -\tfrac{1}{2}\,\epsilon_{ijk} F_{jk}, \tag{57}$$

the six independent components of $F_{\mu\nu}$ are the three components each of the electric and magnetic fields, six independent statistics of the probe-event ensemble; a static source excites only E_i, and the readings of a moving source mix into B_i through the Lorentz structure [1]. The equation of motion of a probe is the covariant form of (46),

$$\frac{dp^\mu}{d\tau} \;=\; q\, F^{\mu\nu} u_\nu\,, \tag{58}$$

u_ν being the four-velocity of the probe, the covariantization given by kinematic inheritance [3], and the static limit returning to (46); the equation holds for macroscopic probes, the velocity of the probe taking its own definite reading (macroscopic trajectories [3]) and the field taking the statistical field quantity. These three equations and the metric readout of §2.4 (time calibration, light calibration, and the mixing of moving sources) are two groups on the same level; the closure of the equations is argued in §3.8. The classical continuous field is the statistical limit at sufficiently large event density, in the same way that macroscopic trajectories are joined from events [3].

When a charge accelerates, the adjustment of the phase field travels outward at the speed of light, which is electromagnetic radiation, and a single share of radiation is the photon itself: the electromagnetic channel has carrier excitations, and its radiation is quantized. The metric structure has no excitation of the propagation operator (§2.6), and gravitational radiation is a collective adjustment of the metric readout, carried by no quantum. Both are non-steady adjustments of sources, one with quanta and one without; the difference lies in whether the channel has carrier excitations, and the algebraic attribution to the two parts of the polar decomposition was given in §1.7.

3.8 The Closure of the Equations

The differential equations of observational spacetime arise from the discrete structural equations of the state space through the statistical readout; the argument for closure proceeds in three steps.

Step 1: the equations hold in the state space for the background quantities themselves. The spectral weight $W(U)$ and the channel phase $a_k(U)$ of a region are the definite values of the background in that region (§1.2), and they obey the structural equations of conservation and propagation: the continuity of shares (24), the stationarity condition of detailed balance (33), the antisymmetric gradient of the phase and charge conservation (§3.3). The subject of the equations is the background quantities; events are only the readout.

Step 2: the readout is an estimate of the background quantities. The sample mean of N_U events in region $U(x)$ has the value of the background in that region as its expectation; the phase of each record is carried away by a photon and the refreshed environment does not return [2], the events are recorded one after another under the same background value, and the sample mean converges to the expectation,

$$\langle f_{\mu\nu} \rangle_{U,N_U} \;=\; F_{\mu\nu}(x) \;+\; O\big(N_U^{-1/2}\big)\,, \tag{59}$$

the fluctuation vanishing as $N_U^{-1/2}$ with the number of events, its coefficient fixed by the correlations of the events, equal to 1 for Poisson counting. The derivative of the observed field with respect to the coordinates is the difference limit of the means of adjacent regions ((11)),

$$\partial_\mu \langle f \rangle \;=\; \langle \partial_\mu f \rangle \;+\; O\big((\delta/L)^2\big)\,, \tag{60}$$

with truncation error of order $(\delta/L)^2$.

Step 3: the statistical field quantities inherit the equations of the background quantities. The equations of the background quantities are linear in the field quantities (§1.7), and a linear equation is preserved term by term under expectation: $F_{\mu\nu} = \partial_\mu \langle a_\nu \rangle - \partial_\nu \langle a_\mu \rangle$ preserves the gradient structure, the source-free set is an identity, and the sourced set has the statistics of the charge current $J^\nu = \langle j^\nu \rangle$ as its source. The statistical field quantities therefore satisfy the Maxwell equations [19] within the two errors, that is, (49) itself; the field strength in the probe equation (58) is the same statistical field quantity, and the two sets compose the complete equations of observational spacetime.

The fork lies in linearity and in the presence of a readout. The color source equation contains the contribution $\rho[a]$ of the carrier itself ((19)) and is nonlinear; color has no sources or probes beyond the color-neutral scale (confinement [1]), the statistical readout has no object, the macroscopic classical equation of color is not given by the readout of this paper, and the observational content of the linearity difference of §1.7 is thereby fixed. The weak equation is linear and closure holds, the solution decays exponentially outside the source and is invisible in the macroscopic readout, its readout residing in microscopic processes: the weak channel opens in processes that change identity [3], and the nuclear potential is read out from nucleon–nucleon scattering and binding energies. The gravitational potential equation is strictly linear in the logarithmic variable (§2.5), closure holds by the same method, all nonlinearity enters through the relation between variable and readout, and the observational equation is written in the readout variable. The classical field equations of the macroscopic readout are therefore exactly two sets, gravity and electromagnetism (§3.6).

The classical equations of observational spacetime hold under two errors: the fluctuation of the event count, $N_U^{-1/2}$, and the discrete truncation, $(\delta/L)^2$ (§1.4); macroscopic event densities put both far below experimental precision. The level of single events has its own exact form: the conservation laws hold as event-by-event balance, the sums of readings before and after a scattering being equal [3]; the equations belong to the statistical limit, the balance belongs to single events, and the two levels are complementary.

3.9 The System of Equations

At this point the equations of fields are all established. The system is organized in three layers following the complete structure of fields (§1.6): discrete structural equations in the state space, differential equations all belonging to observational spacetime, with the conversion between the two layers in §1.4:

State space	
$N_{\rm vac}(U) = N_{\rm vac} - n_{\rm occ}(U)$, $a_k(U)$	region-by-region definition of the fields (3), (4)
$\sum_U(\text{process side}) = \sum_U(\text{background side})$	region-by-region balance (7)
$\Pi_{\rm grav} = \sum_n \lvert\nabla H(n)\rvert^2 \lvert n\rangle\langle n\rvert$	carrying operator (8)
$\lvert A(D,\ell)\rvert \le e^{-\beta D} N(\ell, D)$	propagation bound (6)
$w_{ij}\pi_i = w_{ji}\pi_j$; (33)	detailed balance and discrete stationarity (§2.5)
$\cos(E/v) = \cos(p/v)\cos(m/v)$	lattice dispersion [3]
Readout	
$F_{\mu\nu} = \langle f_{\mu\nu}\rangle_U$	statistical definition (56)
$g_{00} = (v/v_0)^2$, $g_{rr} = g_{00}^{-1}$, $g_{0i} = -4\Phi w_i$	time calibration, light calibration, moving-source mixing (§2.4)
$E_i = F_{0i}$, $B_i = -\frac{1}{2}\epsilon_{ijk}F_{jk}$	component readout (57)
Observational spacetime	
$\partial_t n_{\rm occ} + \nabla\cdot\vec{J} = s$	continuity of shares (24)
$\nabla^2\Phi = 4\pi G\rho_E$; $g_{00} = e^{-2GM/r}$	gravitational source equation and closed form (27), (36)
$\partial_\mu F^{\mu\nu} = J^\nu$	electromagnetism (49) (closure under the statistical average, §3.8)
$(\nabla^2 - M_W^2)a_w = -J_w$; $(\nabla^2 - m_\pi^2)\varphi_N = -g\rho_N$	weak and nuclear (50), (53)
$\nabla^2 a_c = -J_c + \text{self-coupling}$	color (confined, no macroscopic readout, §3.8)
$(\partial_t^2 - \nabla^2 + m_{\rm carrier}^2)a = J$; $(\partial_t^2 - \nabla^2)h^{TT} = 0$	dynamics and waves (55), (41)
$ds^2 = (1+2\Phi)dt^2 - (1-2\Phi)d\vec{l}^{\,2}$	metric (31)
$d^2x^\mu/d\tau^2 + \Gamma^\mu_{\alpha\beta}u^\alpha u^\beta = 0$	geodesic (32)
$dp^\mu/d\tau = qF^{\mu\nu}u_\nu$	probe equation (58)
$\alpha = 4GM/b + 4\pi(GM/b)^2 + O((GM/b)^3)$, $b_c = 2e\,GM$	observables (37), (38)

Each long-range interaction has its equations in all three layers; only two classical sets are complete in observational spacetime, gravity and electromagnetism: the weak and nuclear solutions decay to invisibility in macroscopic regions, and color has no macroscopic sources or probes (§3.8). The equations of observational spacetime correspond item by item to standard physics: the metric and the geodesics agree with general relativity at all orders of the existing tests; the observational Maxwell set and the probe equation are in form the equations of classical electrodynamics. The differences are two. One is the status of the equations: standard physics takes them as fundamental, while here they are the equations satisfied by statistical field quantities, derived from the discrete mechanisms of the state space through the readout (§3.8). The other is the higher orders of gravity: the fork starts at the second order of the spatial component, and it is discriminable (§6).

4 Fields and Light

The source of a field is measured by the phase of exchange, and the exchange share of light is zero: light sits at the zero endpoint of the measure. It sources nothing, while every appearance of a field proceeds through it. The full derivation of the measure, the attribution of the source, and the position of light are the content of this chapter.

4.1 The Measure of Occupation

The full derivation of the occupation measure takes the phase structure of reference [3] and the additivity of the main text as its entire input, in three items.

First, the phase falls into two classes. The phase per cycle splits into the translation share and the exchange share [3],

$$\varphi(k,x) \;=\; k\,\delta(x) \;+\; \frac{m^2}{v(x)\,E}\,, \tag{61}$$

the translation share is the defining property of the characters: $\psi_k(u+\delta) = e^{ik\delta}\psi_k(u)$, a phase increment that is exact, depends on no evaluation, and passes through no convolution spectrum [3]; the exchange share is produced by formation exchange, evaluated on the convolution spectrum through the even channels [1]. Occupation is the use of carrying, and the use occurs in exchange (§1.2): consumption goes only through the exchange share, and the translation share costs nothing.

Second, the uniqueness of the unit of measure. Consumption counts in shares, and the number of shares is proportional to the amount of exchange phase, not to the number of exchanges. Of the two candidates only one survives,

$$\Delta n\Big|_{\text{by number}} \;\propto\; 1\,, \qquad \Delta n\Big|_{\text{by phase amount}} \;\propto\; \frac{M}{v}\,, \tag{62}$$

counting by number, a composite standing wave obeys one and the same single-cycle transfer [3], one exchange per cycle, so occupation is independent of the number of components, contradicting additivity (20); counting by phase amount, the exchange phase of a composite per cycle is M/v, the binding shares counted in automatically [3], so occupation is proportional to M, which is exactly (21).

Third, the dependence of occupation on motion. The exchange share of a single free standing wave per cycle is $m^2/(vE)$ and the cycle rate is v, so the consumption rate is

$$\frac{m^2}{vE}\cdot v \;=\; \frac{m^2}{E} \;=\; m\cdot\frac{m}{E}\,, \tag{63}$$

reducing to m at rest; in motion it is diluted by m/E, and m/E is the ratio of proper time to coordinate time [3]: occupation occurs proper cycle by proper cycle, proper time slows, and the coordinate consumption rate drops by the same factor.

4.2 The Attribution of the Source

With the measure fixed, the attribution of the source in the bound and unbound cases follows.

A density identity gives the covariant identity of the source. In the continuous approximation $E^2 = p^2 + m^2$, for a free stream of number density n, the rate density of the exchange share is

$$n\,\frac{m^2}{E} \;=\; n\Big(E - \frac{p^2}{E}\Big) \;=\; \rho - 3p\,, \qquad p \;=\; \frac{n\,p^2}{3E}\ (\text{isotropic})\,, \tag{64}$$

and the rate density of the translation share is $n\,p^2/E = 3p$. Source and pressure each receive an algebraic identity: the occupation density of an unbound free stream equals the trace of the energy-momentum tensor, and the pressure equals one third of the translation-share rate density. Occupation is a count, the trace is a Lorentz scalar, and the covariant identity fits the counting nature.

Bound and free are distinguished by compensation. The total exchange phase of a composite per cycle is M/v [3], the free parts of the components total $\sum_a m_a^2/(vE_a)$, and the difference is borne by the constraining interactions (collisions and the exchange of potentials); the exchange share of the constraints is exactly this difference,

$$\sum_a \frac{m_a^2}{E_a} \;+\; \big(\text{exchange share of constraints}\big) \;=\; M\,, \tag{65}$$

so the total occupation is proportional to M; this is the microscopic mechanism of the statement in the main text that "inside a bound body the pressure terms and the binding energy cancel in the total" (§2.2). Unbound free streams have no such compensation, and their occupation density is the trace $\rho - 3p$.

4.3 The Source of Light

That the source of light is zero follows from the splitting of phase. The phase per cycle of light is entirely translation share,

$$\frac{E}{v} \;=\; \frac{p^2}{vE} \;+\; 0 \qquad (E = p)\,, \tag{66}$$

the exchange share is zero [3], occupation is zero, and freely propagating light sources nothing. The same conclusion has three independent routes: the continuous limit of the exchange share m^2/E as $m \to 0$; the tracelessness of the electromagnetic field; the literal statement of reference [3]. Response and sourcing separate from here on,

$$\text{response} \;\propto\; \frac{E}{v}\,, \qquad \text{sourcing} \;\propto\; \frac{m^2}{vE}\,, \tag{67}$$

response acts on the whole phase, the phase of light is entirely translation share, both parts register, and the deflection is twice that of slow trajectories (§2.4); sourcing goes only through the exchange share. An excitation can respond fully while sourcing nothing, and light is exactly that. The source of a free stream takes the trace $\rho - 3p$ (64), and light is its zero endpoint; (12) at $m = 0$ gives the same conclusion at the level of the basic units.

4.4 Light Calibration

Light calibration gives the spatial readout of fields. Slow excitations register only the position dependence of the cycle rate, light registers both, and the deflection is twice the bending of slow trajectories (§2.4); the spatial metric is defined by the propagation of light,

$$g_{rr}(x) \;=\; \left(\frac{v_0}{v(x)}\right)^2 \;=\; g_{00}^{-1}(x)$$

(§2.4). The spatial part of the metric is therefore not an independent quantity but the reading of light calibration; the tests of the spatial component (light deflection, time delay, the shadow) are all samplings of this reading by the events of light (§6).

4.5 Light and Statistics

Every appearance of a field proceeds through photon events. The carrier of records is uniquely the photon [2]; the field strength is the statistics of the photon-event ensemble, the readings of single events (energy, momentum, polarization) giving $E_i = F_{0i}$ and $B_i = -\frac{1}{2}\epsilon_{ijk}F_{jk}$ through the ensemble average (§3.7); a gravitational wave produces no events, it only changes the statistics of photon events, and what the interferometer reads is the latter (§2.6). The carrier of statistics and the carrier of records are one and the same: the motion of reference [3] appears through photon records as a sequence of events, the fields of this paper appear through the same carrier as the statistics of events, and the observational faces of the two papers share a single readout channel.

5 The Phenomena of Fields

§1 defined fields, §2 and §3 established the equations, and §4 gave the position of light. This section places that structure among the puzzles of fields, from action at a distance to the classical limit; each puzzle is resolved by one part of the structure.

5.1 Action at a Distance

The gravity of the Sun acts on planets hundreds of millions of kilometers away, and a magnet attracts iron filings across vacuum; changes of a source arrive at finite speed, the gravitational wave of the binary neutron star merger and the gamma-ray burst arriving 1.7 s apart after traveling for over a hundred million years [14].

Step 1: the redistribution of shares proceeds site by site, the flow $\vec{J}$ of the continuity equation (24) is nearest-neighbor transfer, and none of the three origins of conservation contains a direct term across a distance (§2.2). The amplitude reaching metric distance D within ℓ cycles obeys the propagation bound (6),

$$|A(D,\ell)| \;\le\; e^{-\beta D}\, 2^{\ell} \;=\; \exp\!\big[-\beta\big(D - \ell \ln 2/\beta\big)\big]\,,$$

arrival faster than $\ln 2/\beta$ per cycle is exponentially suppressed, and outside the light cone there is only an exponential tail; once a source is established, the distribution inside the front approaches step by step the stationary stock of the transfer chain (§2.2), and the static field is the completed shape of the transfer.

Step 2: the propagation bound fixes only an upper limit, and the actual speed is given by the dispersion of the zero-mass channels. Gravitational waves and light share one and the same dispersion (40), $E = \pm p$ holds exactly on the lattice, propagation is one site per cycle, and the vanishing of the speed difference is an identity (§2.6). The ratio of the arrival interval of the two messengers to the travel time gives the direct upper bound on the speed difference,

$$\frac{\Delta v}{c} \;\sim\; \frac{c\,\Delta t}{D} \;=\; \frac{1.7\,\mathrm{s}}{\sim 4\times 10^{15}\,\mathrm{s}} \;\sim\; 10^{-16}\,,$$

and the measured bound of 10^{-15} [14] tests exactly this identity. Action at a distance has no corresponding term in the structure; a field is the stock shape of the transfer chain.

5.2 The Photoelectric Effect

Light shines on a metal surface and photoelectrons leave one by one, the counting rate proportional to the intensity, whether they leave decided by the frequency alone [5, 22]; under faint radiation a detector responds click by click, counting share by share; in a long exposure the image develops photon by photon, while the instrument reading of the field strength is continuous.

Step 1: a single readout has no magnitude, only the two outcomes of occurring or not occurring, the effective overlap taking only the values 0 and 1 [3]; the share nature of events comes from the two-valuedness of records and has nothing to do with any division of the field.

Step 2: each cycle allows at most one exchange, the probability of a single record is the electromagnetic coupling α [3], the weight of landing in a region is the squared modulus of the amplitude (the Born rule [25, 1]), and the expected number of events factorizes as

$$\langle N_U\rangle \;\propto\; \alpha\cdot v\,T\cdot |A|^2$$

(T the observation time), the response rate, the cycle rate, and the intensity each occupying one factor; the photocurrent being proportional to intensity is the readout of the last factor. The energy of a single event is given by event-by-event balance, the sums of readings before and after the scattering being equal [3], the share of the photon equaling the sum of the maximal ejection energy and the work function,

$$E_e^{\max} \;=\; E_\gamma \;-\; W\,,$$

when the share falls below the work function no record forms, and the threshold depends on frequency alone; the maximal ejection energy is linear in frequency and independent of intensity, the strength of the illumination changing only the number of events.

Step 3: the phase of each record is carried away irrevocably by the photon, the environment refreshing does not return [2], no correlation between events remains through the environment, and the fluctuation of the count vanishes with the scaling of independent increments,

$$\frac{\Delta N_U}{\langle N_U \rangle} \;\propto\; \langle N_U \rangle^{-1/2}\,,$$

the coefficient fixed by the correlations of the source itself; the sample mean converges to the field quantity at the same rate ((59)); the development of a lensed image [6] is the spatial form of this convergence, the estimation error of the contrast of the image decreasing with the number of events as $\langle N \rangle^{-1/2}$.

Continuity and discreteness therefore do not conflict: continuity is the limit of statistics, discreteness is the nature of events; the wave–particle duality of fields is the two levels of the readout, the same structure as the double slit of motion [3].

5.3 The Localization of Field Energy

The orbit of a binary shrinks and energy leaves the source; where the radiation arrives, a detector receives energy; between source and detector, the attribution of the energy has no reading.

Step 1: the gradient of the radiation-reaction potential (44) does work on the source, and the time average of the power, after two integrations by parts, reduces to a positive-definite term beyond total derivatives,

$$\Big\langle \frac{d^5 Q_{ij}}{dt^5}\,\frac{dQ_{ij}}{dt} \Big\rangle = -\Big\langle \frac{d^4 Q_{ij}}{dt^4}\,\frac{d^2 Q_{ij}}{dt^2} \Big\rangle = +\Big\langle \dddot{Q}_{ij}\,\dddot{Q}_{ij} \Big\rangle\,,$$

which is $P = cG\langle \dddot{Q}_{ij}\dddot{Q}_{ij}\rangle$ of (45), positive definite.

Step 2: $\sum_{\text{out}} R = \sum_{\text{in}} R$ holds exactly event by event [3], and the energy reception of the detector passes through no statistics.

Step 3: between source and detector there are no events and no reading of energy; the fluctuation weight of the regions the wave passes through is unchanged (the sum of squared proper spacings is $\delta l^2 \sum_i (1 + h_i) = 3\delta l^2$, tracelessness removing every order, §2.6), the regions acquire no occupation from the passage of the wave, and the energy balance closes only between source and detector, by conservation.

The energy density of the gravitational field has no coordinate-independent value in general relativity; in observational spacetime there are no fields (§1.6), the pointwise energy density accordingly has no carrier, and the difficulty is a direct corollary of the main thesis (§2.6).

5.4 The Graviton

A single-photon detector responds click by click to faint electromagnetic radiation, one share each time; when a gravitational wave passes, the interferometer reads a continuous shift of the fringes, at strain 10^{-21} [31]; no instrument has ever recorded gravitational radiation share by share.

Step 1: the metric structure has no excitation of the propagation operator [1], an excitation able to carry records must be charge-neutral, precisely frequency-matched, and massless so as to escape, the triple sieve leaving only the photon [2] (§2.6); the quantum that the quantization program envisages for gravity has no corresponding excitation in this framework.

Step 2: electromagnetic radiation counts share by share, the probability of a single record is the electromagnetic coupling α [3], and the response rate of a single-photon detector is the readout of this probability; gravitational radiation has no share-by-share structure, what the interferometer reads is the shift of the statistics of photon events, the fringes of the two arms accumulate from photon events, the passing wave adjusts the metric readings and translates the statistical distribution, and the strain reading is the amount of that translation.

Step 3: the phase reading of the fringes is the mean of the event distribution, the standard error of the mean of N events is $N^{-1/2}$ ((59)), converted into arm-length difference and strain,

$$\delta(\Delta L) \;\sim\; \frac{\lambda}{2\pi}\,N^{-1/2}\,, \qquad h_{\min} \;\sim\; \frac{\lambda}{2\pi L}\,N^{-1/2} \;\sim\; 10^{-21}$$

($\lambda \sim 1\,\mu\text{m}$, $L \sim 4\,\text{km}$, $N \sim 10^{20}$ within the integration time), the shot-noise term [23] being the fluctuation term of the statistical readout, its value depending on the measurement scheme; on the noise curve of the instrument the fluctuation term of (59) is a measured quantity, and its magnitude is exactly the strain scale of gravitational-wave observation.

Share-by-share counting and the statistical shift are the form the radiation-quantization difference of §1.7 takes at the instrument.

5.5 The Classical Limit

The readouts of the preceding four subsections all carry statistical structure. The everyday field, however, is definite: the needle of an electrometer is steady, the equations of engineering electromagnetism hold precisely, and planetary ephemerides are computed with classical gravity.

Step 1: the deviation of a region reading from the statistical limit is $\langle N_U\rangle^{-1/2}$ ((59)), the number of events is given by the share-by-share arrival of energy, the energy received by a region divided by the share of a single photon,

$$\langle N_U\rangle \;=\; \frac{I\,S\,T}{E_\gamma} \;\approx\; \frac{1\,\text{mW/cm}^2 \times 1\,\text{cm}^2 \times 1\,\text{s}}{2.3\,\text{eV}} \;\approx\; 3\times 10^{15}$$

(I the intensity, S the area, T the duration, E_γ the share of a single visible-light photon), and the fluctuation

$$\langle N_U\rangle^{-1/2} \;\approx\; 2\times 10^{-8}\,,$$

below the resolution of instruments.

Step 2: the truncation of differential equations against differences is of order $(\delta/L)^2$ (§1.4), the lattice spacing is one step of the cycle, $\delta = 1/v \approx 8\times 10^{-19}$ m, and a laboratory scale $L \sim 1$ m gives

$$(\delta/L)^2 \;\sim\; 6\times 10^{-37}\,,$$

twenty-nine orders of magnitude below the fluctuation; the scale of planetary ephemerides is the astronomical unit, where the truncation is deeper still. The difference between the readings of the statistical field quantities and the continuous classical field is composed of the two errors, the fluctuation of this example at the 10^{-8} level, and the statistical field quantities with their equations are indistinguishable in readings from the continuous classical field (§3.8).

The classical field is the readout of the statistical limit, in the same way that macroscopic trajectories are joined from events; the quantum–classical divide of motion lies in two ratios, mass suppressing the spreading rate and records suppressing the event interval [3]; the statistical–classical divide of fields lies in two errors, the fluctuation of the event count and the discrete truncation.

6 Predictions

The nonlinearity of general relativity comes from the self-sourcing of the field: the energy of the gravitational field enters its own equation. The potential of this paper is strictly linear in the balance variable, the field carries no source of its own, and gravity has no energy-carrying carrier (§1.7, §2.6).

1. The second-order coefficient of the spatial metric component is 2, against 3/2 in general relativity, guaranteed jointly by the locking $g_{00}g_{rr} = 1$ and the proper constitutive relation. The observable is the second-order light-deflection coefficient, 4π here against $15\pi/4$ in general relativity, a relative deviation of $+1/15$, at solar grazing 11.7 versus 10.9 μas. Existing first-order precision measurements are not contaminated by this deviation: at solar grazing the second-order difference leaks into the γ reading by $(\pi/8)(GM/b) \approx 8 \times 10^{-7}$, more than an order of magnitude below the current best bound 2.3×10^{-5} [30, 9]; the threshold for judging the second order is astrometry at the μas level.

2. The third-order coefficient of the time metric component is $-4/3$, against $-3/2$ in general relativity; both sets are taken in isotropic-type gauge, and with the first two orders aligned the third-order difference is physical. Third-order effects near the solar surface are at the $(GM/r)^3 \sim 10^{-17}$ level, far beyond current observation, and impose no present constraint.

3. The potential is bounded below, $\Phi \geq -1$, and g_{00} is positive everywhere: strong-field objects have no horizon at finite radius, and the redshift grows without bound with depth; the critical impact parameter of light is $b_c = 2e\,GM$, and the shadow diameter is 4.6% larger than in general relativity, a spherically symmetric value with rotational corrections separate, the formation of the shadow further requiring the optical conditions of the interior. Current horizon-scale imaging is compatible: the shadow-deviation measurements for Sgr A*, $\delta = -0.08 \pm 0.09$ and -0.04 ± 0.10 [10], stand at 0.9–1.4σ from the $+0.046$ of this paper. The ringdown is a quantity awaiting judgment: the main ringdown mode is determined by the photon-sphere structure [11], and by the geometric-optics correspondence the orbital frequency and the instability rate of the photon sphere here are both $3\sqrt{3}/(2e) = 0.956$ of the general-relativity values, that is, the frequency and the damping rate of the fundamental mode are lower by the same 4.4%, of the same order as the existing bounds of a few percent on the frequency deviation of that mode; the judgment requires the rotating solution and the perturbation equations on the curved background of this model, and echo-like signals are the further criterion.

4. The source of gravity is the energy density without pressure, while the static effective source of general relativity is $\rho + 3p$. Outside bound bodies the two are equivalent, and existing solar-system and binary tests are unaffected; the forks are two: the interior structure of compact stars, where the maximum mass under the same equation of state awaits the interior structure equation; and the source of unbound free streams, where the measure of occupation gives the trace $\rho - 3p$, including that free-streaming light sources nothing (§2.2, §4.2).

5. Gravitational waves are not carried by quanta: the metric structure has no excitation of the propagation operator, and the only radiation carrier passed by the triple sieve is the photon, which does not belong to the metric direction (§2.6); a share-by-share record of gravitational radiation, if observed, refutes this item.

6. The trace component of the radiation field, the breathing mode, is zero, its origin being the relaxation of the trace component on the weight level rather than gauge symmetry; existing polarization observations favor pure tensor modes [13].

7. The linear dispersion of the two zero-mass channels, light and gravitational waves, is exact on the lattice and carries no discrete correction (§2.6); the existing strong bounds on linear dispersion from high-energy photon arrival times [12] are consistent with this, and it is discriminable against programs predicting Planck-scale dispersion; the massive channels retain a dispersion deviation of order $(\delta/L)^2$ [3].

7 Conclusion

A field is a state of the background in the state space, produced and maintained by the cycles of standing waves. The cycles of a standing wave occupy the spectral weight of the region they sit in and accumulate channel phase along dwell segments, and the two correction quantities extend into position-by-position distributions through conservative redistribution; the equations of the distributions are consequences of conservation and propagation, occupation giving the Poisson equation and the closed-form metric, phase giving the unified static equation and the wave forms. The difference between the two types of field comes from the algebras of the two halves of the polar decomposition. The modular part is non-negative, and the interaction it carries is always attractive; the involution exchanges the two members of a conjugate pair, the charges come in two signs, and the interactions come with both directions; whether the carrier carries its own charge separates linear from nonlinear.

The only channel by which the state of the background enters observational spacetime is the record. The field strength is the statistics of the probe-event ensemble, with photons carrying the records, and the metric is read out by time calibration and light calibration; the equations of the statistical field quantities close term by term through linearity, into the classical Maxwell equations and the geodesics. The classical field equations of the macroscopic readout are exactly two sets, gravity and electromagnetism: the weak and nuclear solutions decay to invisibility in macroscopic regions, and color has no macroscopic sources or probes. The two errors of the statistics, the fluctuation of the event count and the discrete truncation, are both far below experimental resolution, and the classical appearance is thereby settled.

The status of fields is the same as the status of motion [3]: in spacetime there is no motion, only the records of motion; in spacetime there are no fields, only the records of the background quantities. The source is a recorded process, located where the standing-wave cycles correct the background; the source terms on the right-hand sides of the equations are process-side readings, not inputs by induction. The observed field is the event statistics of the state of the background.

Data availability

No data sets were generated or analysed during this study; all results follow from analytical derivation.

Conflict of interest

The author declares no conflict of interest.

A Exclusion of Alternative Forms

The structural choices of the main text are given by the structural derivations; the alternative forms listed, independently of the derivations of the main text, contradict observation on their own, and the structural derivations and the observational exclusions meet item by item.

1. The linearity of the source is given in §2.1 by additivity through the Cauchy equation, and in §3.2 carried by the additivity of accumulation standing wave by standing wave. If occupation were counted by the squared-fluctuation convention ($n_{\text{occ}} \propto E^2$), counting a composite standing wave as a whole gives M^2, the potential is nonlinear in mass, contradicting the linearity of the Newton potential; counting by components gives $\sum_i m_i^2 \approx M\, m_p$ (nucleon-dominated), the source changes with composition, iron and water samples of equal mass fall differently, contradicting the 10^{-13} precision of the equivalence-principle tests.

   The same for electromagnetism: if accumulation were proportional to q^2, the positive and negative charges of a neutral atom would not cancel, leaving a net source, contradicting the 10^{-21} precision of the tests of the electric neutrality of matter. The squared forms are excluded by observation in both cases.

   The other source of the squared form is misreading the participating weight of the energy-scale formula as occupation: the participating set and occupation are two different quantities (§1.2), the participating weight scales with the square of the characteristic energy, while occupation is linear in energy.

2. The exchange form is given in §2.2 by conservation and the additive weight shift of the χ_0 direction. If shares propagated freely along straight lines instead of exchanging region by region, what crosses a sphere would be a flux rather than a stock, the potential would decay as $1/r^2$, contradicting the Newton potential; redistribution must be region-by-region exchange.

3. The structural reason why process-universal numbers take no part in localization is given in §1.2: every region runs one and the same condensation process. If C^* or N_{vac} took local values over spectral intervals, the local energy scale of the vacuum would vary exponentially with spectral position (the fluctuation weight $|\nabla H(n)|^2$ decays as $1/n^2$ with the mode label, and partial sums over spectral intervals are exponentially uneven in position), contradicting the uniformity of the vacuum; only the correction quantities take part in localization.

4. The structural proof that the trace does not radiate is in §2.6: the trace is a probability-level conserved count, its evolution driven by a dissipative generator with real negative spectrum, supporting no traveling wave. If the transverse trace component (the breathing mode) were a free radiation mode, polarization observations already constrain it strongly [13], and the radiation modes are only the two transverse traceless ones. The language of exchange-particle spin does not appear in this paper; the structure of signs and polarizations is carried by the polar decomposition and the conservation laws.

5. The structural derivation of the exponential form is given in §2.5 from the two physical inputs of detailed balance and the uniform walk. The observational criterion for exclusion is single: the perihelion precession fixes the second-order coefficient of g_{00} at $+2$ (§2.5).

   Coordinate-level models write the transfer of shares on a rigid coordinate lattice network, with flat operators acting directly on the transferred quantity; the three representative models each take a combination of g_{00} as the flat harmonic variable (writing $y = g_{00}$, harmonicity outside the source means the combination is proportional to $1 - \text{const} \cdot x$, first order normalized to -2):

Model	Flat harmonic combination	Expansion	Second-order coefficient
Hopping type (pile-up by lingering)	$y(1-y)=2x$	$1-2x-4x^2+\cdots$	-4
Gradient type	$y^2=1-4x$	$1-2x-2x^2+\cdots$	-2
Site-wise equal-sharing type	$y^{3/2}=1-3x$	$1-2x-x^2+\cdots$	-1

The second-order coefficients are all negative, contradicting $+2$: the transfer cannot be written on a rigid coordinate lattice and must be written on the network of events (§2.5).

The literal transcription of site-by-site equal sharing is excluded at the coordinate level, and the balance combination $(1-\Phi)^2$ it would write is excluded along with its origin; its closed form $\Phi = 1-\sqrt{1+2x}$ agrees with the exponential solution at first and second order, has third order -2, and has an extremal-type boundary of vanishing surface gravity at $r = 2GM/3$; the network derivation from the equilibrium measure (§2.5) gives the exponential form.

The parametrization of the measure itself is constrained by the same criterion. With conductance and stationary measure $\kappa \propto g_{00}^a$, $\pi \propto g_{00}^b$, the spherically symmetric stationarity condition $\kappa\,(n/\pi)'\,r^2 = \text{const}$ of (34) becomes $g^{\,a-b-1}\big[g + b(1-g)\big]\,g'\,r^2 = \text{const}$ ($g = g_{00}$, $n \propto 1-g$), separation of variables integrates to $G(g) \propto x$ with $G'(g) = g^{\,a-b-1}[g + b(1-g)]$; expanding G at $g = 1$ and normalizing the first order,

$$g_{00} \;=\; 1 - 2x - 2\,(a-2b)\,x^2 + O(x^3)\,, \tag{68}$$

the second-order coefficient is $4b-2a$, observation requires $4b-2a = 2$, the constraint line $a = 2b-1$. The uniform walk $(1,1)$ lies exactly on the line, and $G(g) = \log g$ makes the exponential solution exact at all orders; the naive alternatives all lie off the line, their second-order coefficients read directly from (68): independent per-share jump rates $(0,-1)$ give -4 (the proper-level counterpart of the coordinate-level hopping type), interface-fixed rates $(1,0)$ give -2 (the gradient type), complete mixing $(2,1)$ gives 0, the jump rate uniform in proper time $(3/2,1)$ gives $+1$ (closed form $g_{00} = (1-x)^2$), all excluded, and the uniform walk on the event network is the only microscopic model among those listed that passes observation. The line holds more than one point: the third-order coefficient varies along it, $(1,1)$ giving $-4/3$ and $(3,2)$ giving -4, so third-order observation can judge the measure itself; $(-1,0)$ and $(1,1)$ give one and the same equation at all orders, the exponential solution does not invert to the coefficients of the coarse network, and the network is given by the construction of §2.5.

The localization of the measure is likewise excluded. If the measure $\log^2 W$ of each share of fluctuation varied with the local weight, the $\log^2 N_{\text{vac}}$ of (5) would become $\log^2 W$, g_{00} would acquire the factor $(\log W/\log N_{\text{vac}})^2$, and the expansion gives a second-order coefficient 2.37 (shift $4(\varepsilon^2+\varepsilon)/(1+2\varepsilon)^2$, $\varepsilon = 1/\log N_{\text{vac}}$), contradicting the perihelion precession; that the measure takes its vacuum value (§1.2) is independently forced by observation.

6. The locking is given in §2.4 jointly by light calibration and the invariance of the spectral spacing. If the spatial component were an independent dynamical field instead of being locked by $g_{00}g_{rr} = 1$, its proper length would need an operational definition beyond light calibration; the spectral spacing of the substrate is not altered by matter [1] and cannot serve as a variable proper ruler, while the ruler of matter (the sizes of bound structures) gives the same reading as light calibration (§2.4), so an independent field has neither a bearer nor a readout different from light calibration. The locking of the spatial component is thereby the consistency of the readout. What is excluded is a static independent ruler field; the dynamical transverse traceless components are the direction-resolved readout of light calibration itself (§2.6) and are not in this class.

B Order-by-Order Expansion of the Closed Form

This appendix gives the order-by-order comparison of the closed-form metric with the Schwarzschild solution, and converts the second-order fork into observables independent of coordinate gauge, supplying the derivations for the comparison table of §2.5 and item 1 of §6.

The solution (36) expands in $x = GM/r$:

$$g_{00} = e^{-2x} = 1 - 2x + 2x^2 - \tfrac{4}{3}x^3 + \cdots , \qquad g_{rr} = e^{2x} = 1 + 2x + 2x^2 + \tfrac{4}{3}x^3 + \cdots .$$

The Schwarzschild solution in isotropic coordinates is

$$g_{00}^{\mathrm{GR}} = \Big(\frac{1-x/2}{1+x/2}\Big)^2 = 1 - 2x + 2x^2 - \tfrac{3}{2}x^3 + \cdots , \quad g_{rr}^{\mathrm{GR}} = \big(1 + \tfrac{x}{2}\big)^4 = 1 + 2x + \tfrac{3}{2}x^2 + \tfrac{1}{2}x^3 + \cdots .$$

The two expansions agree in full at first order and in g_{00} at second order; the deviations lie in g_{rr} at second order (2 versus 3/2) and in g_{00} at third order ($-4/3$ versus $-3/2$). The existing solar-system tests (light deflection, time delay, perihelion precession) are sensitive to the first order and the second order of g_{00}, where the two are indistinguishable.

The second-order comparison in observables. The orbit of light depends only on the ratio $F^2 = g_{rr}/g_{00}$; under the locking $F^2 = 1/g_{00}^2 = e^{4x} = 1 + 4x + 8x^2 + \cdots$; in the isotropic coordinates of general relativity $F^2 = (1 + x/2)^6(1 - x/2)^{-2} = 1 + 4x + \frac{15}{2}x^2 + \cdots$. With the invariant impact parameter $b = L/E$ (the ratio of orbital conserved quantities, an observable), the deflection angle

$$\alpha \;=\; 2\int_0^{u_0} \frac{du}{\sqrt{F^2(u)/b^2 - u^2}} \;-\; \pi \tag{69}$$

integrates order by order: the first-order coefficient is that of F^2 (both 4); the second-order coefficient obeys the parametrized formula

$$\alpha_2 \;=\; \pi\Big[\,2 + 2\gamma - \beta + \tfrac{3}{4}\,\delta_2\Big] ,$$

where γ, β are the first- and second-order PPN parameters and δ_2 is defined by the second-order coefficient $\frac{3}{2}\delta_2$ of g_{rr}: this paper has $\gamma = \beta = 1$, $\delta_2 = 4/3$, giving 4π; general relativity has $\delta_2 = 1$, giving $\frac{15}{4}\pi$ (verified independently by numerical integration to 10^{-5}; the general-relativity value agrees with the known result). The gauge dependence cancels in the ratio and the invariant parameter, and the second-order spatial fork appears in pure observables: 4π versus $\frac{15}{4}\pi$, relative deviation $+1/15$; (37) is the second-order truncation, with remainder $O((GM/b)^3)$. The same ratio F^2 determines the strong-field critical impact parameter $b_c = \min_u F(u)/u$: $F = e^{2GMu}$ gives $u^* = 1/2GM$, $b_c = 2e\,GM = 5.437\,GM$, against $3\sqrt{3}\,GM = 5.196\,GM$ in general relativity (the numerical minimization checked exactly against the Schwarzschild value); $g_{00} = e^{-2x} > 0$ for all $r > 0$, with no finite boundary.

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